# Object detection

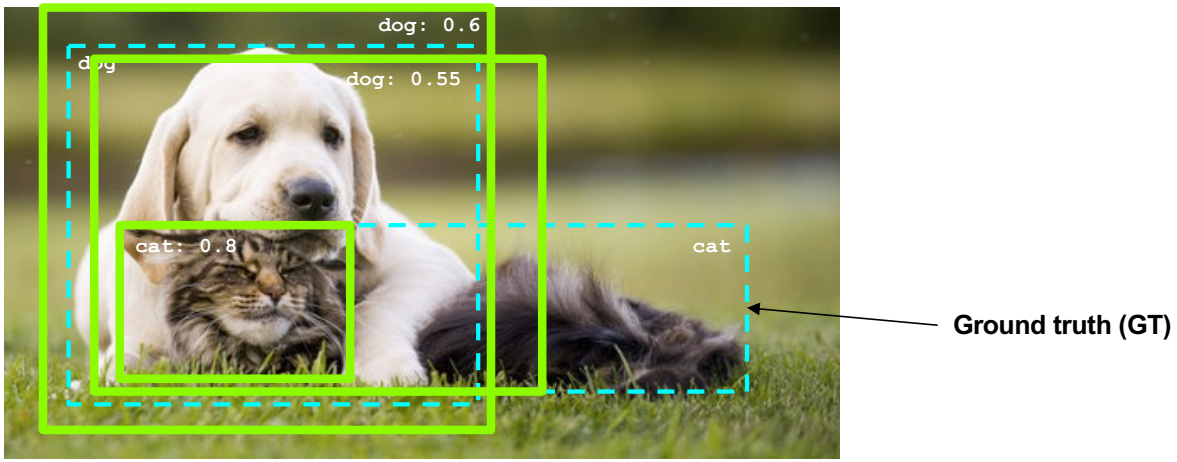

Image source

# Outline

- Task definition and evaluation
- Two-stage detectors:
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - Mask R-CNN
- Single-stage and multi-resolution detectors
- Other detectors

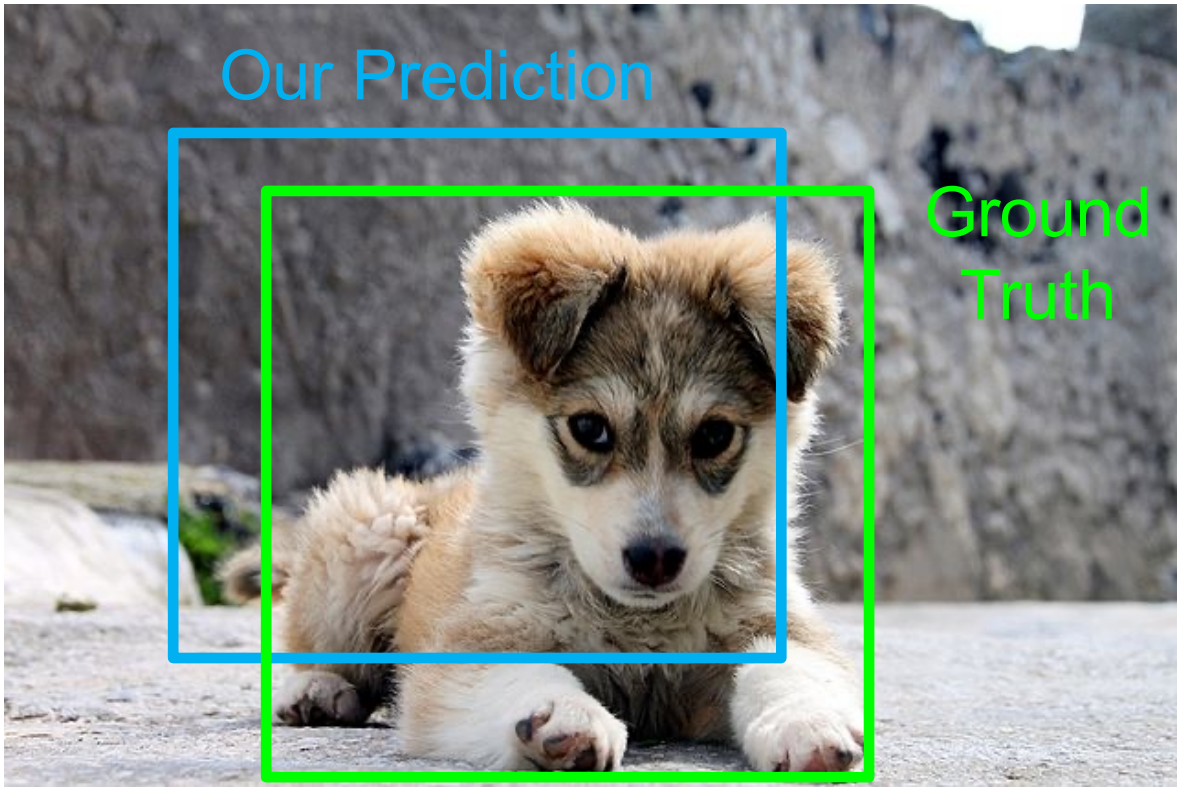
# Task definition and evaluation

- At test time, predict bounding boxes, class labels, and confidence scores
- For each detection, determine whether it is a true or false positive
  - But how?



# Comparing Boxes: Intersection over Union (IoU)

How can we compare our prediction to the ground-truth box?




Source: J. Johnson

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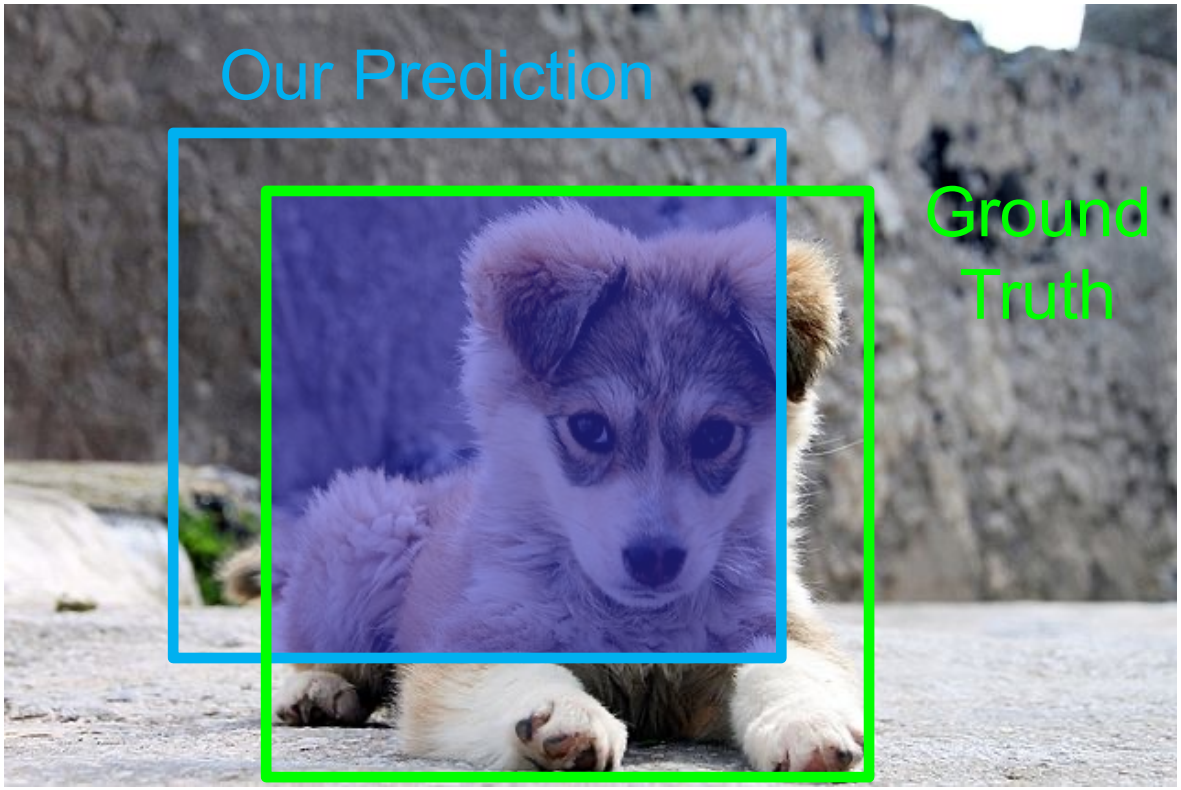


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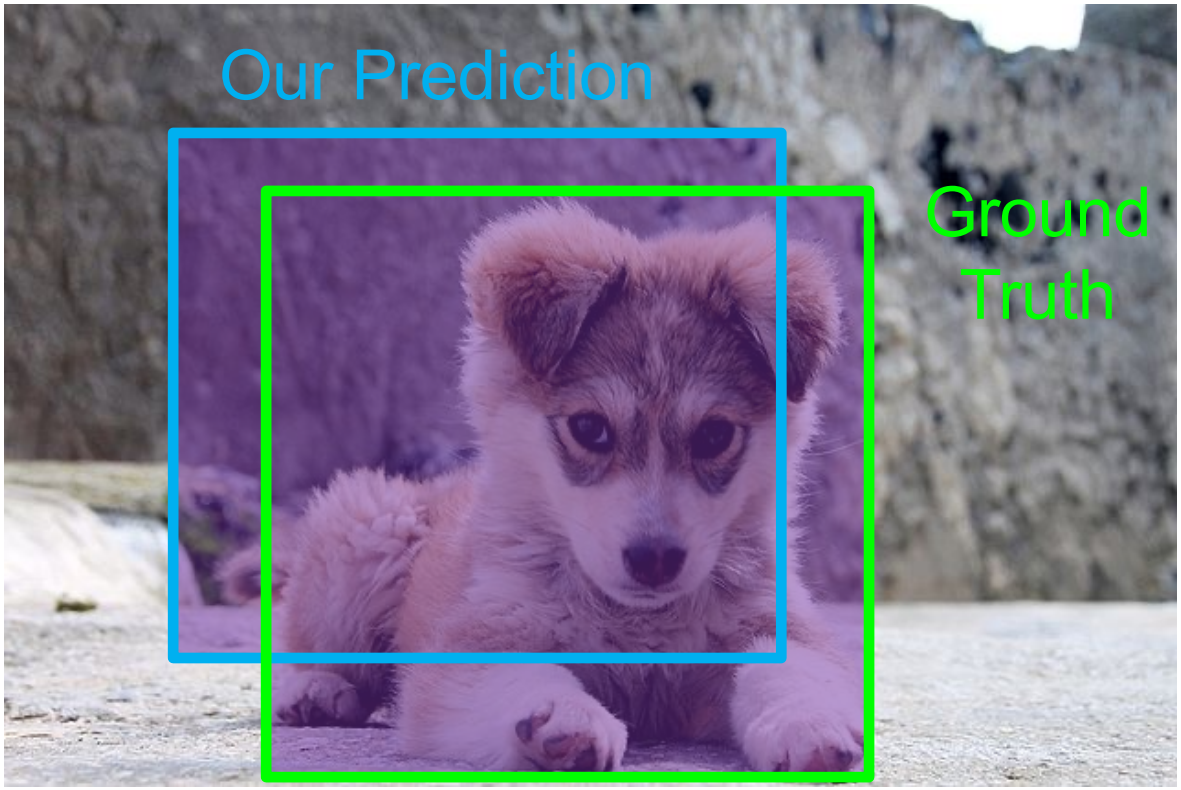


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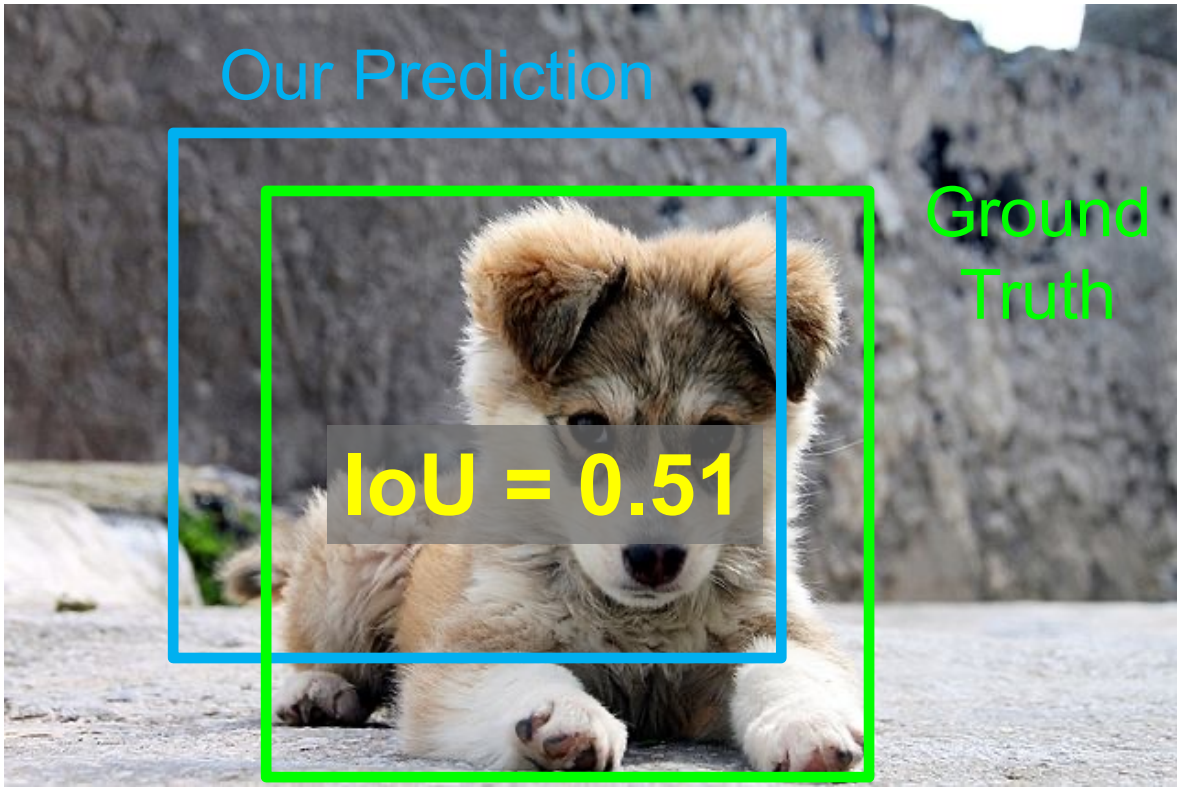


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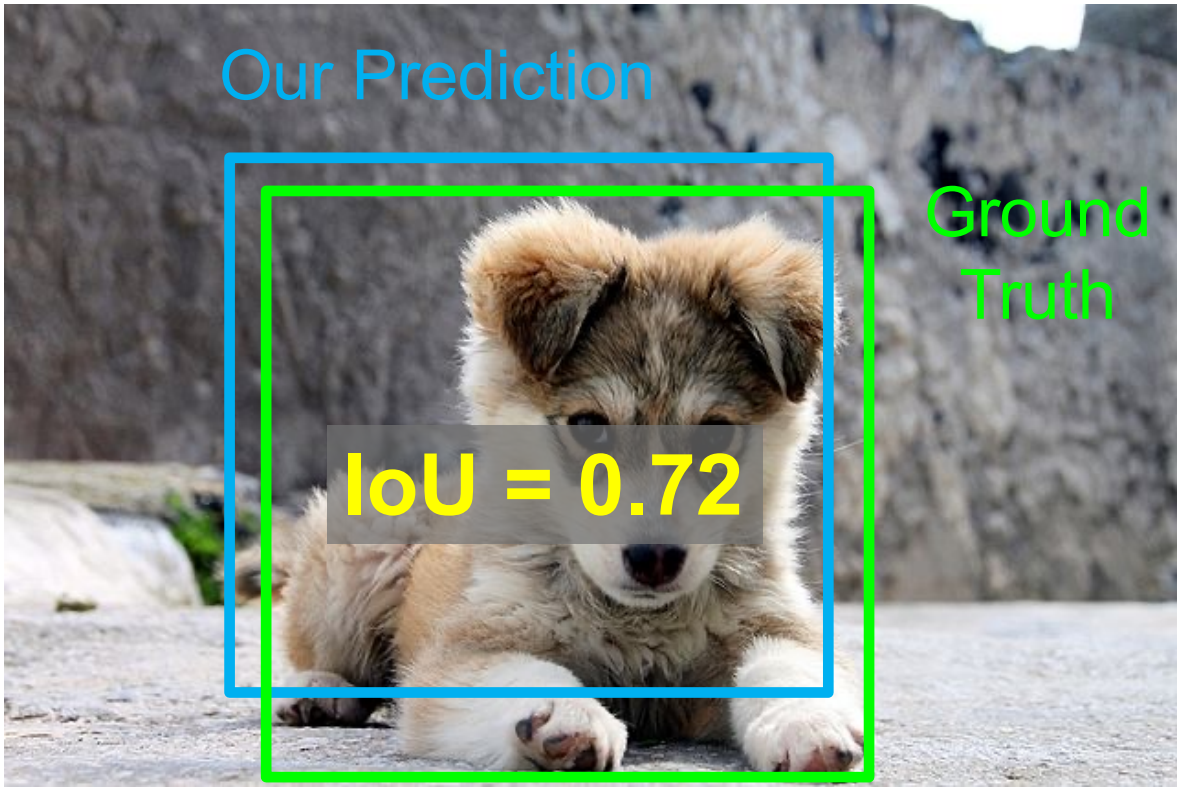


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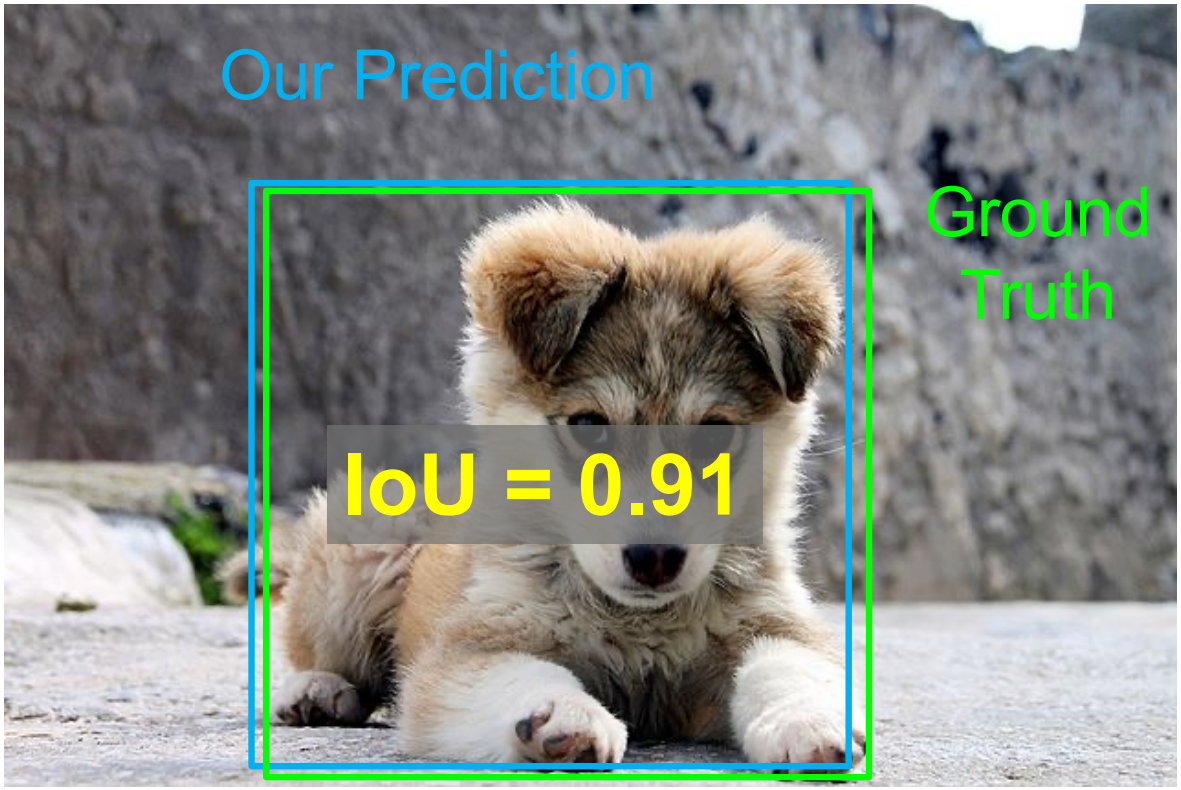


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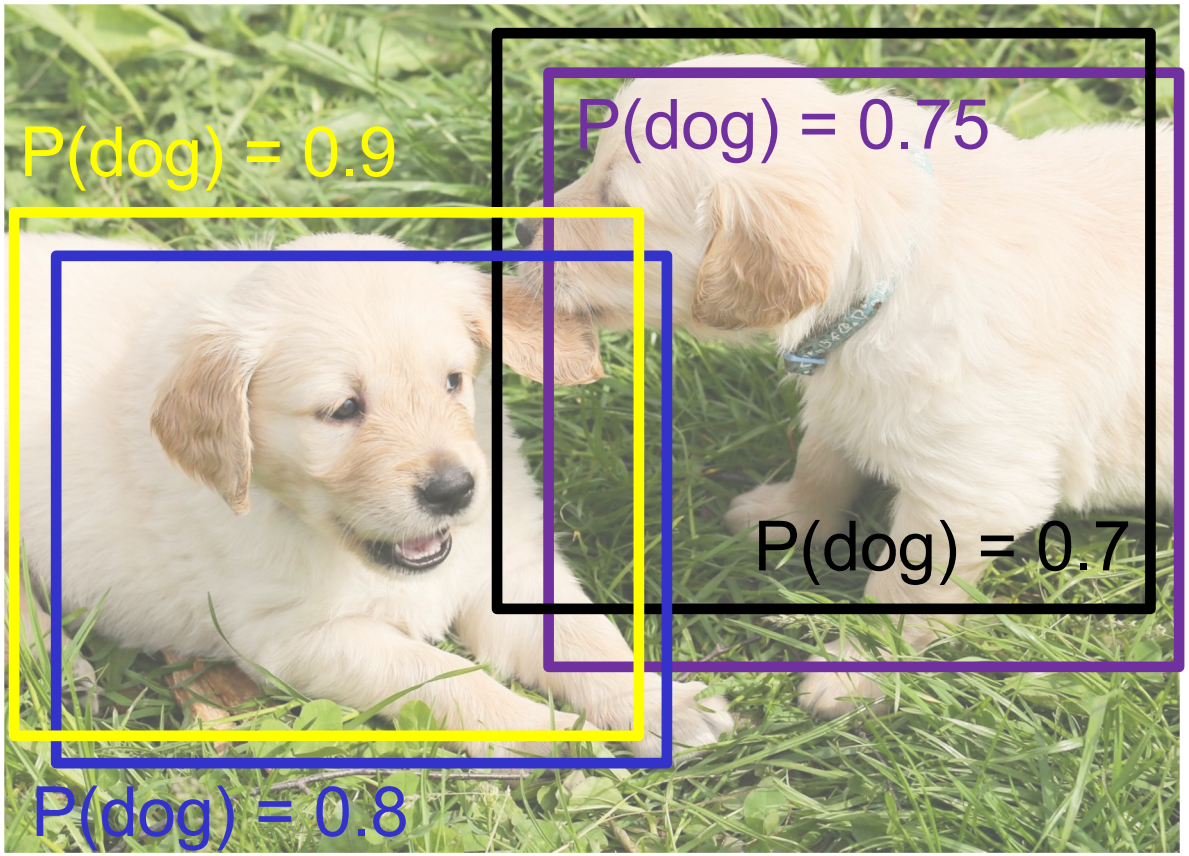
# Non-maximum suppression

Problem: Detectors often output many overlapping detections

Solution: Post-process raw detections using Non-Maximum Suppression (NMS)

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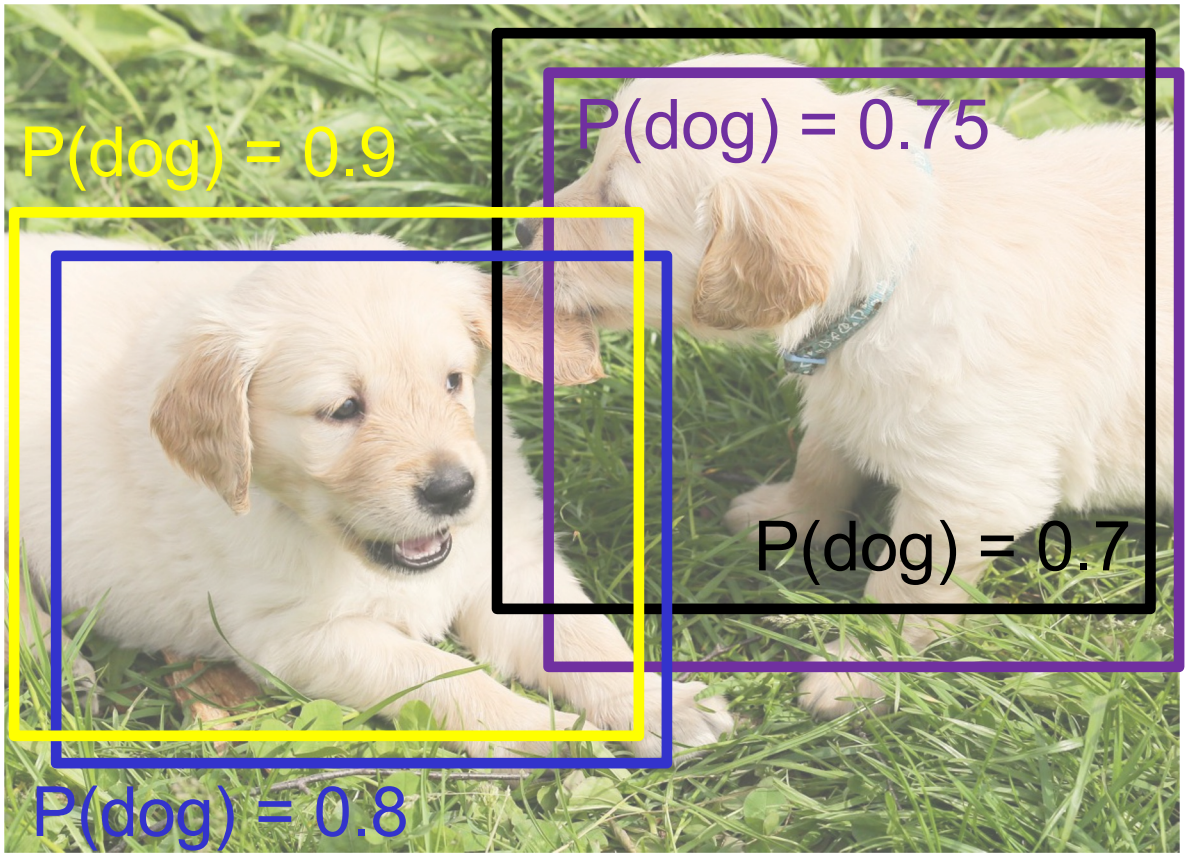


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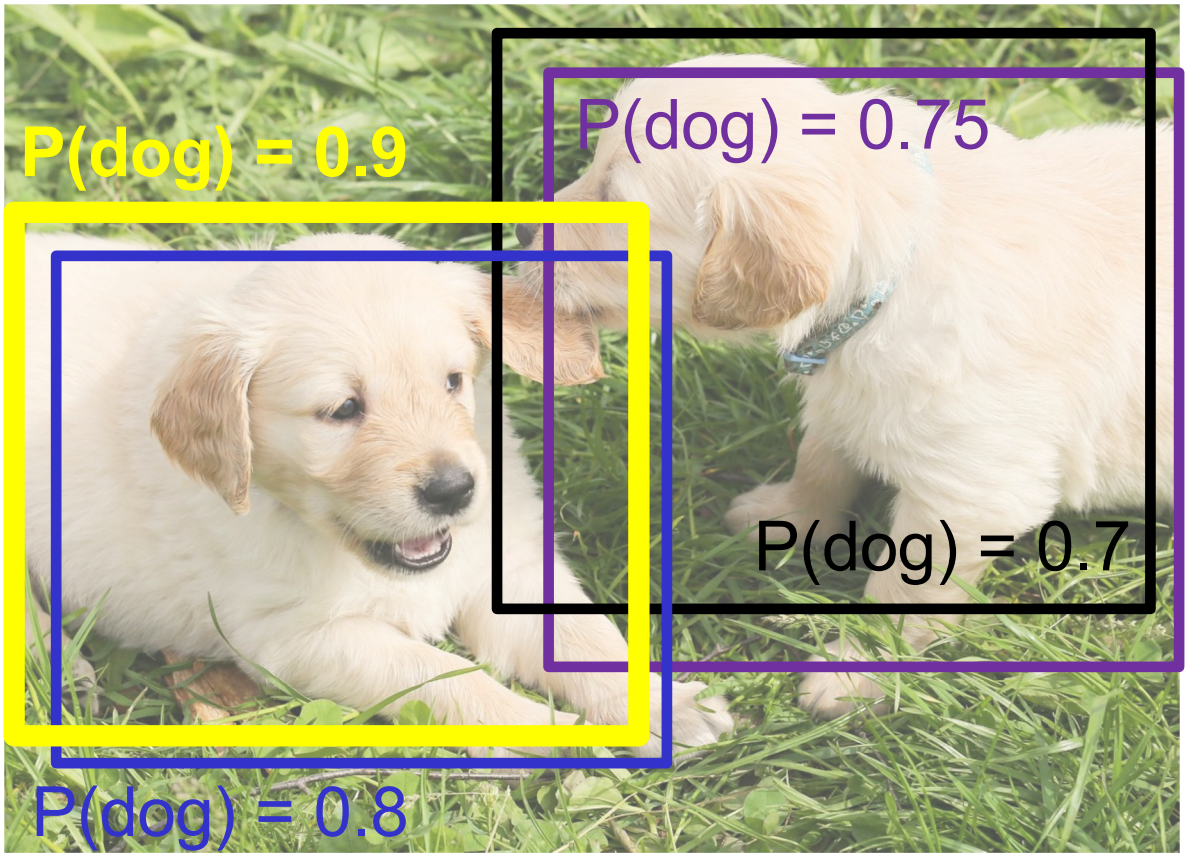


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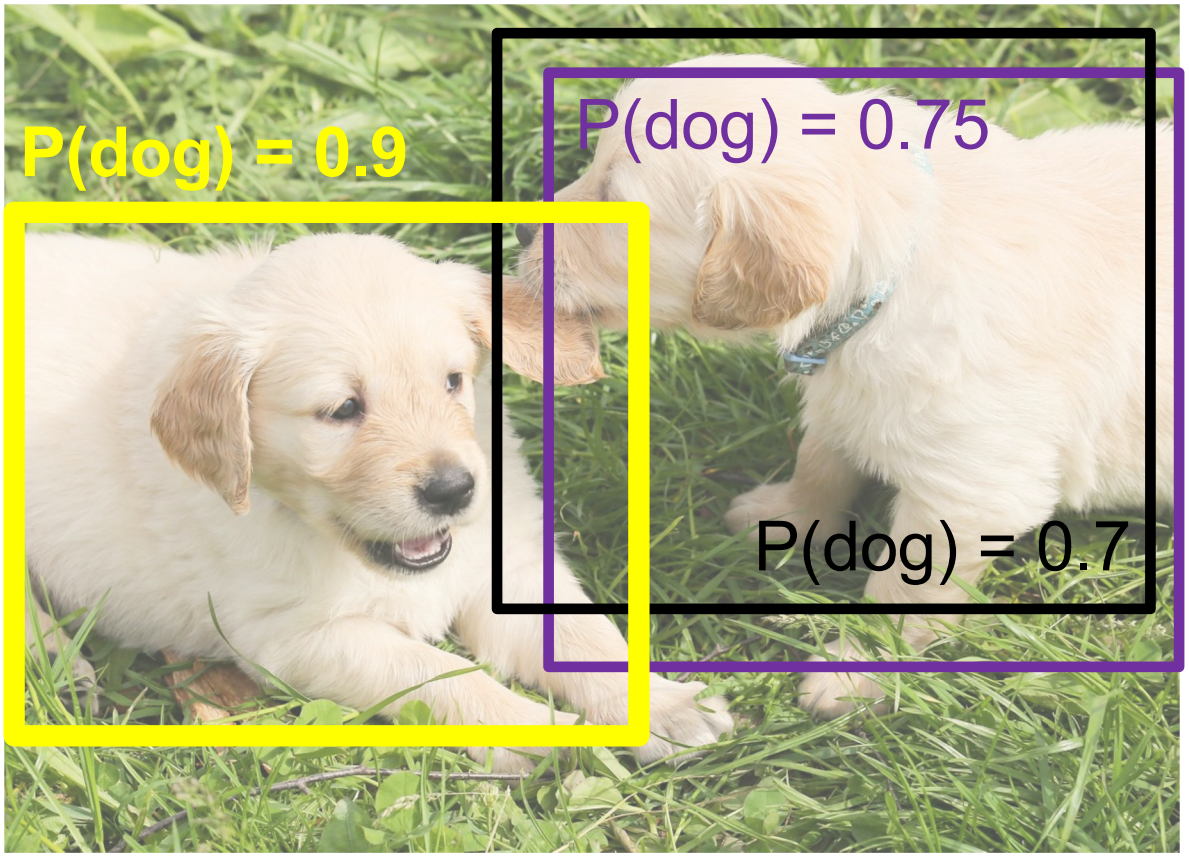


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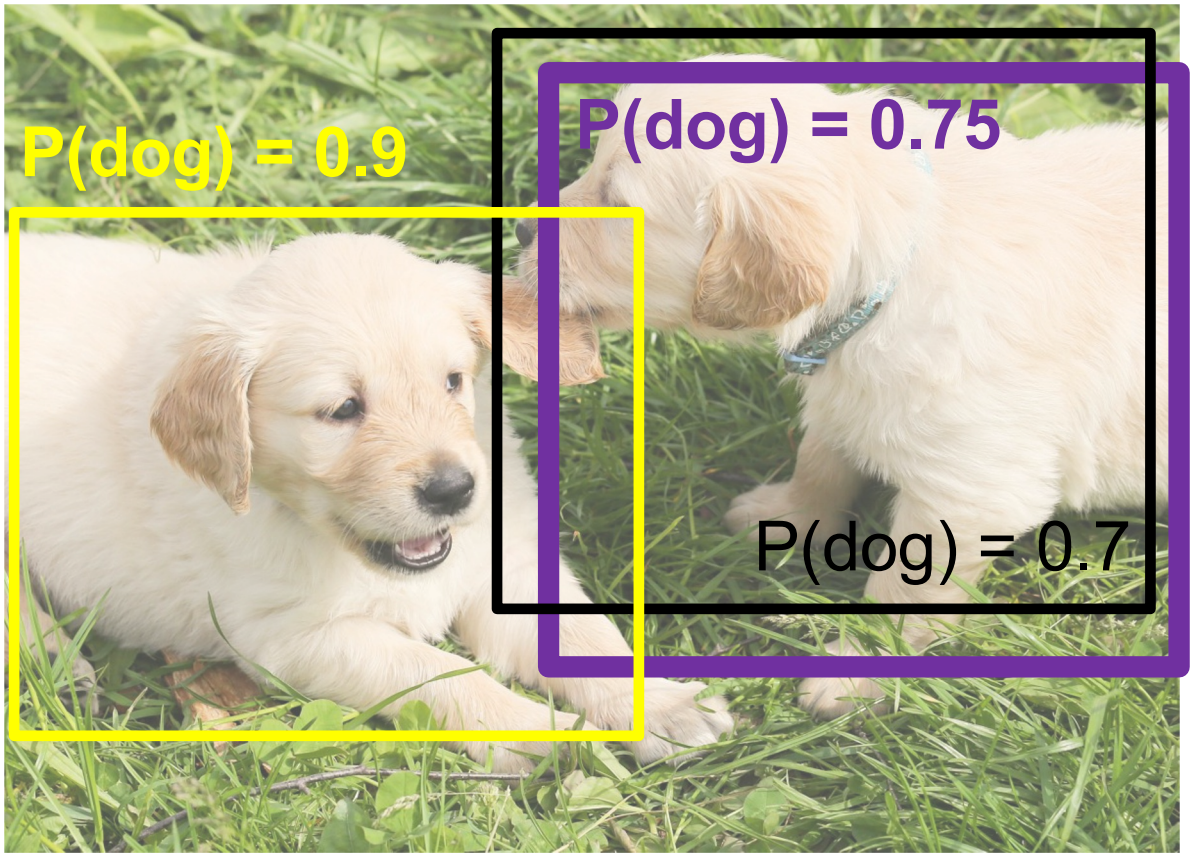


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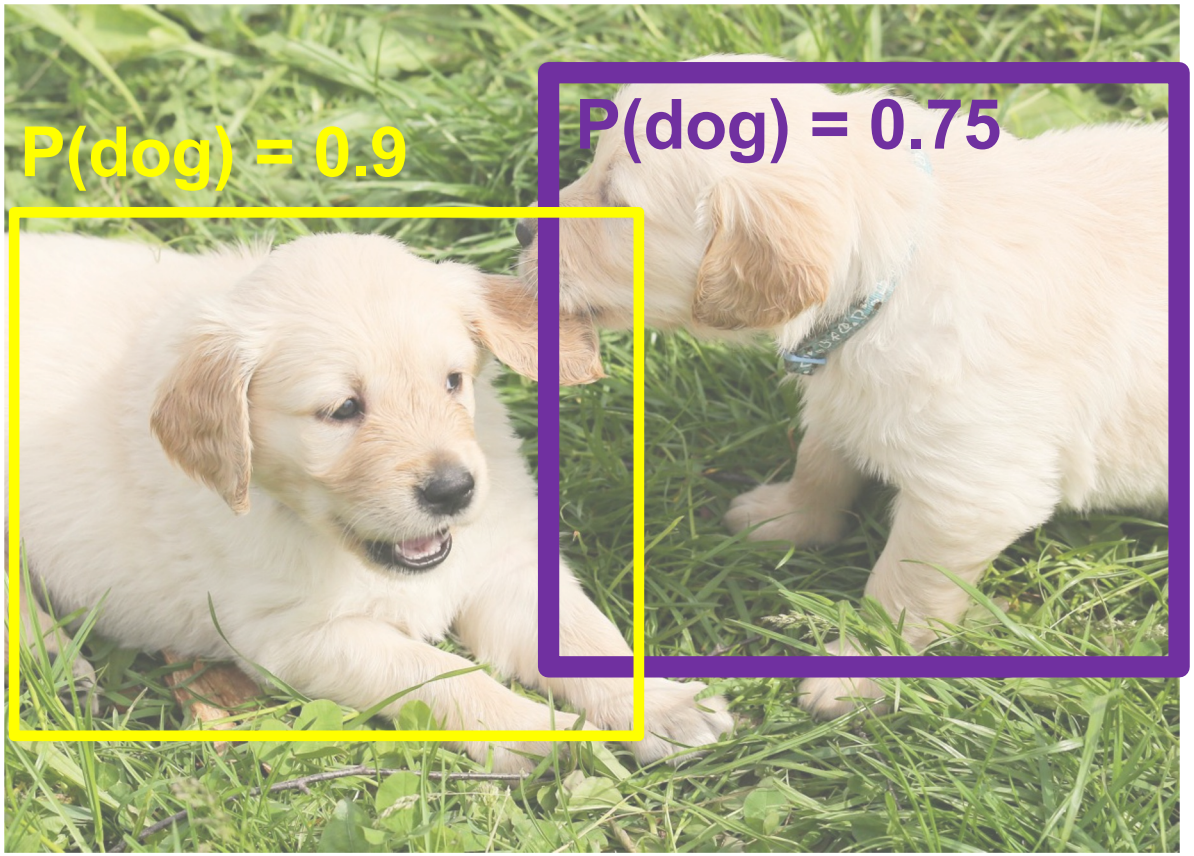


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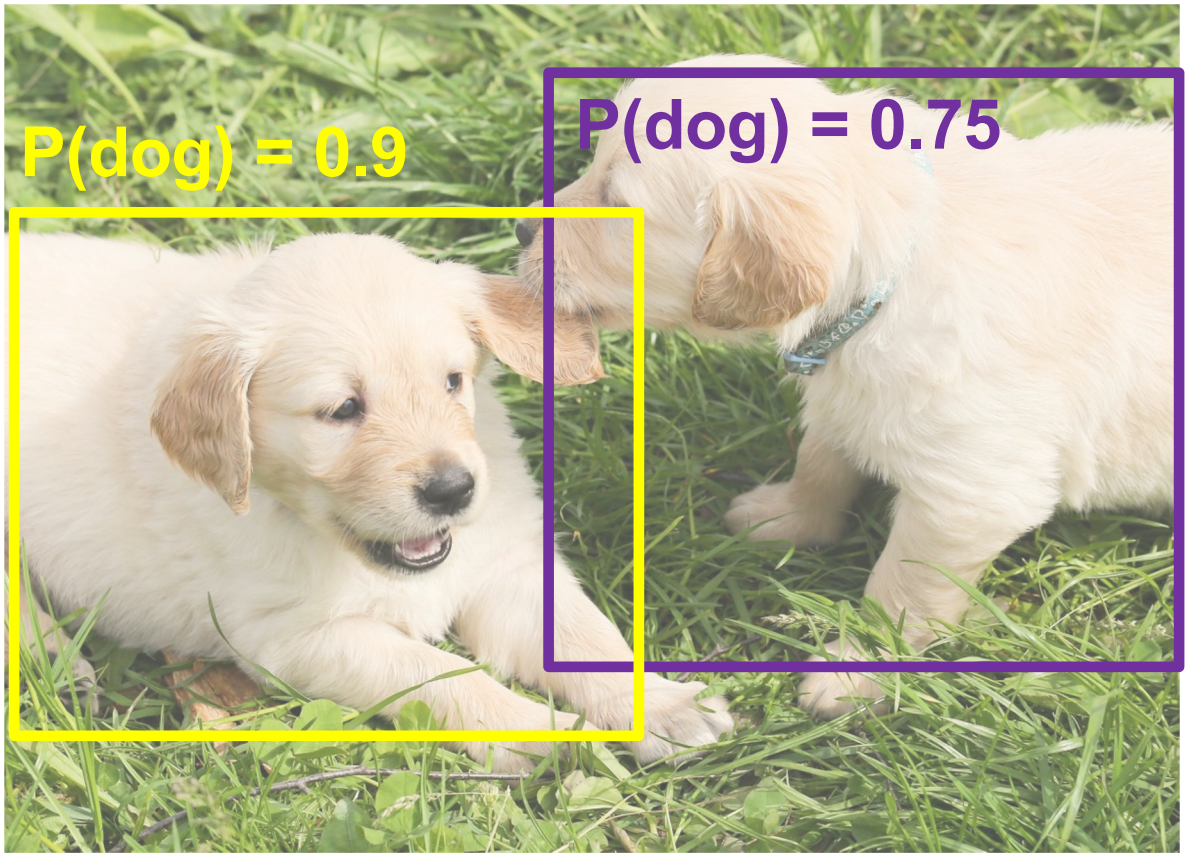


Source: J. Johnson

# Non-maximum suppression: Limitations

# Non-maximum suppression: Limitations

How would NMS do on an image like this?

- It will eliminate “good” boxes when objects are highly overlapping

Source: J. Johnson

# Evaluating object detectors

1. Run object detector on all test images (with NMS)
2. For each category, compute the **Precision vs. Recall Curve** and **Average Precision (AP),** or area under the Precision vs. Recall Curve
3. Average the APs of all categories to obtain **mean AP**, or **mAP**

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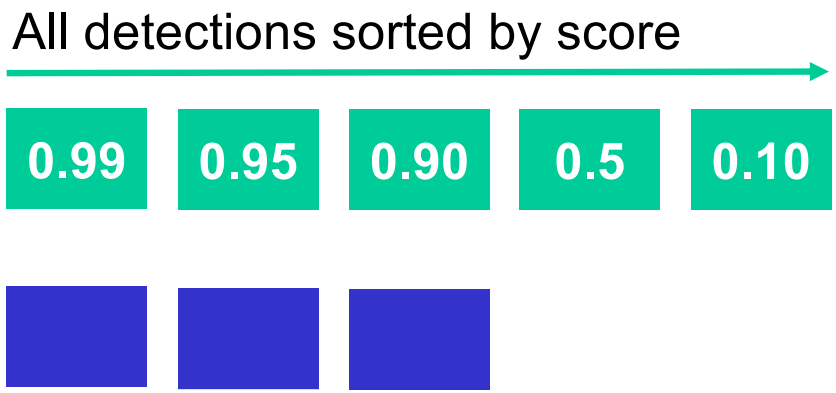


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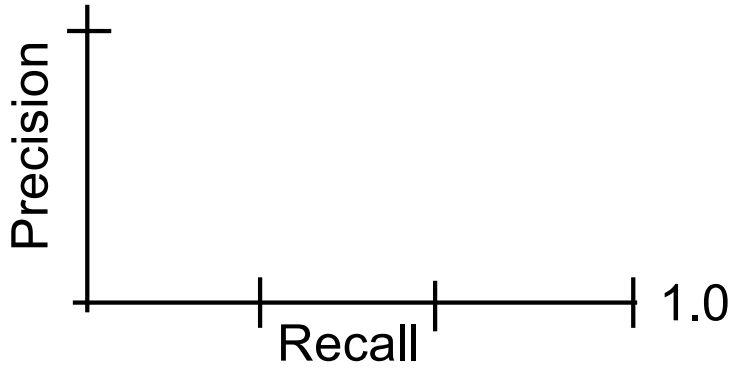


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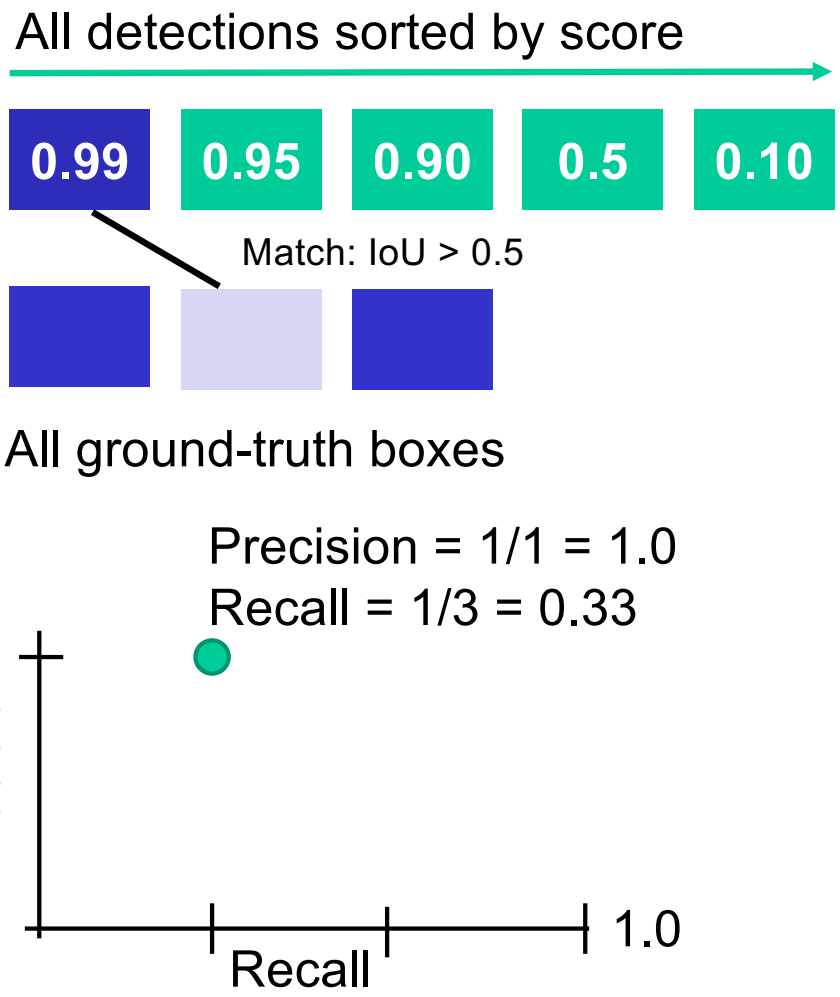

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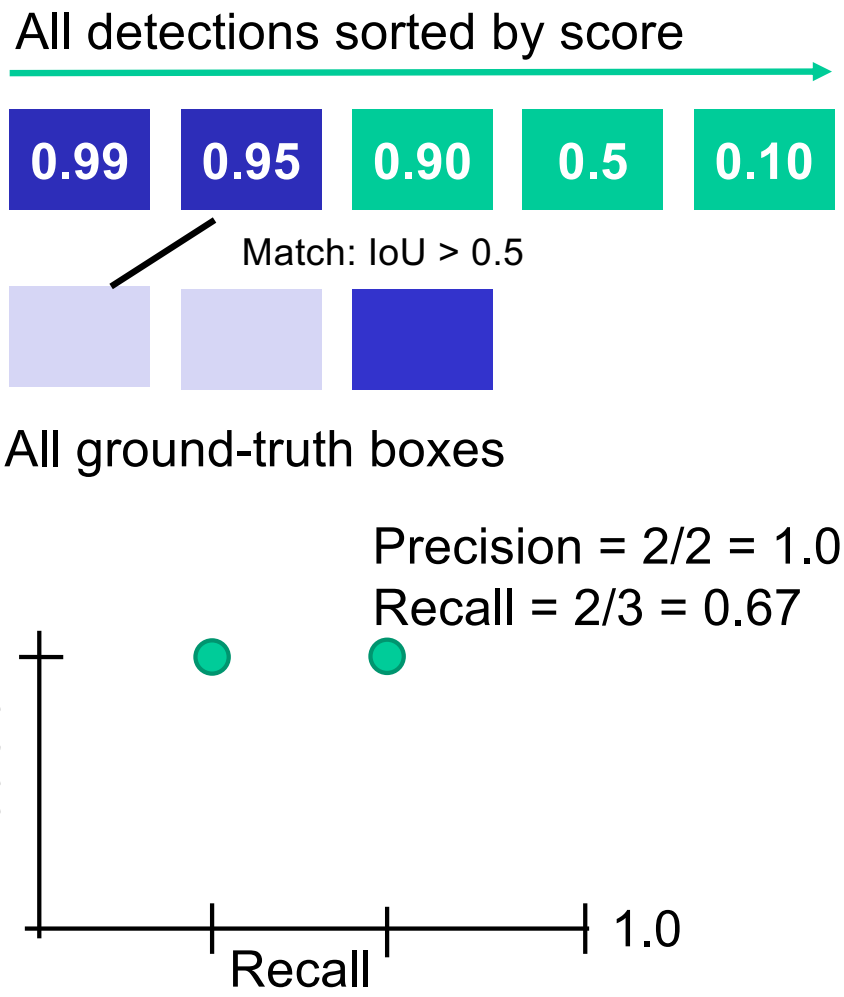

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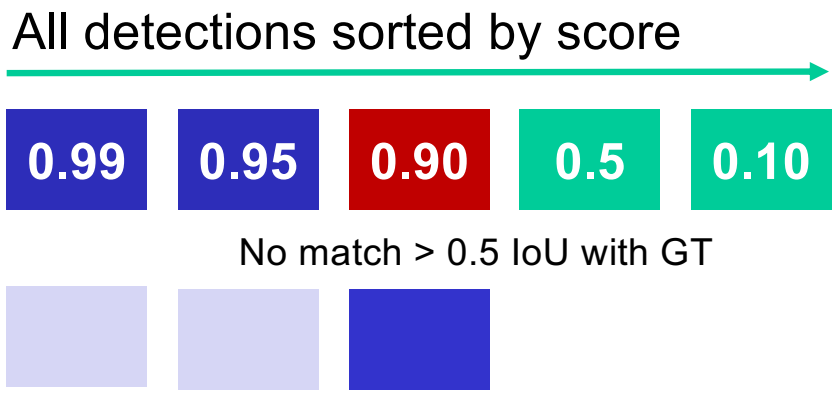


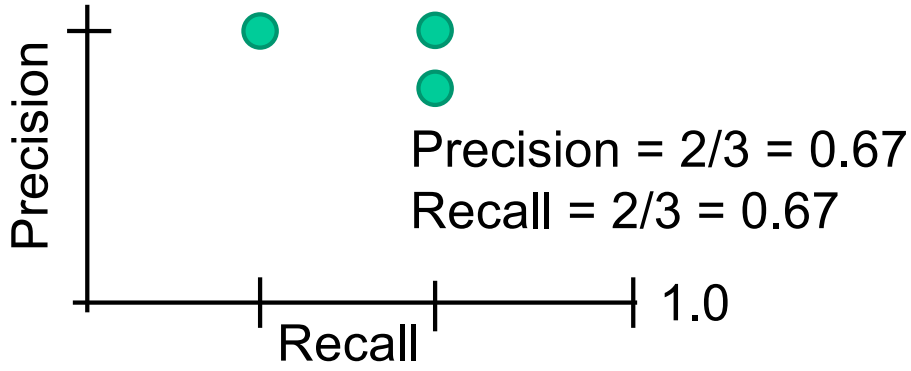

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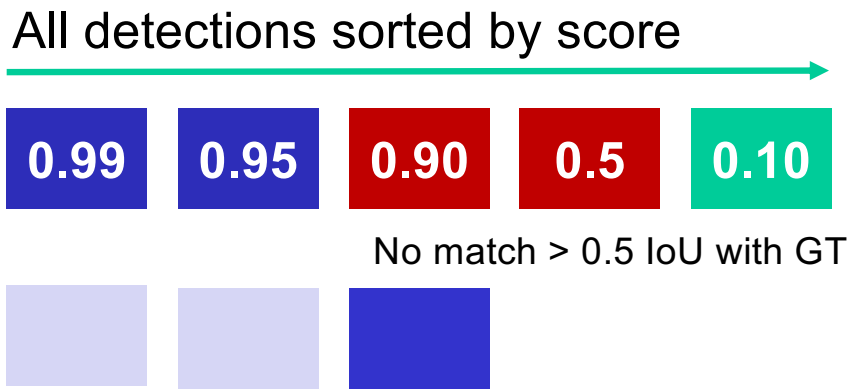


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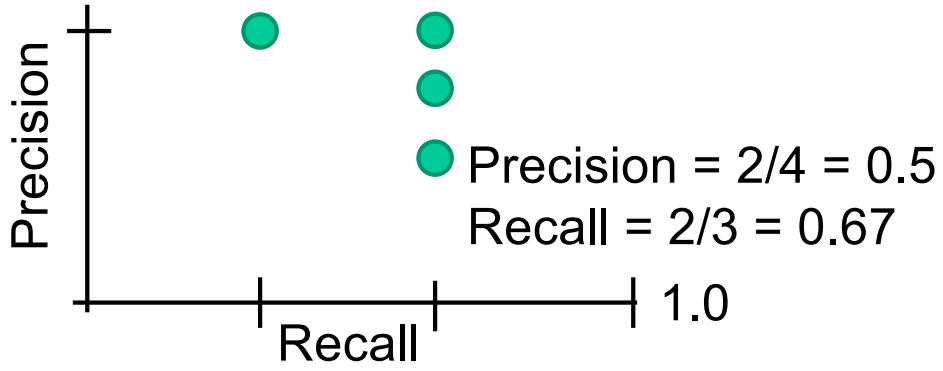


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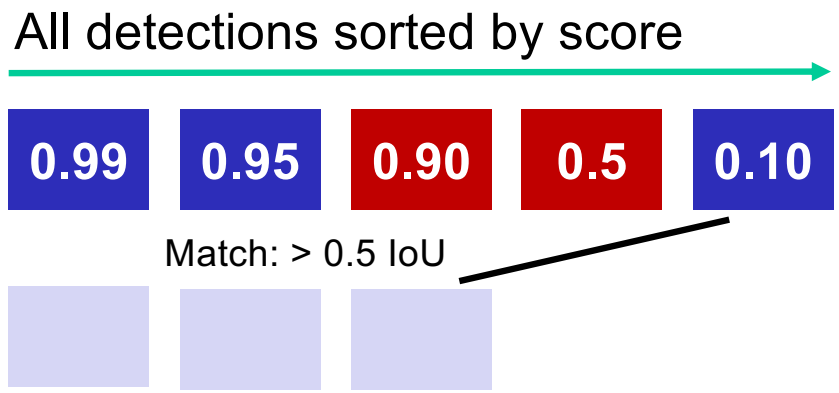


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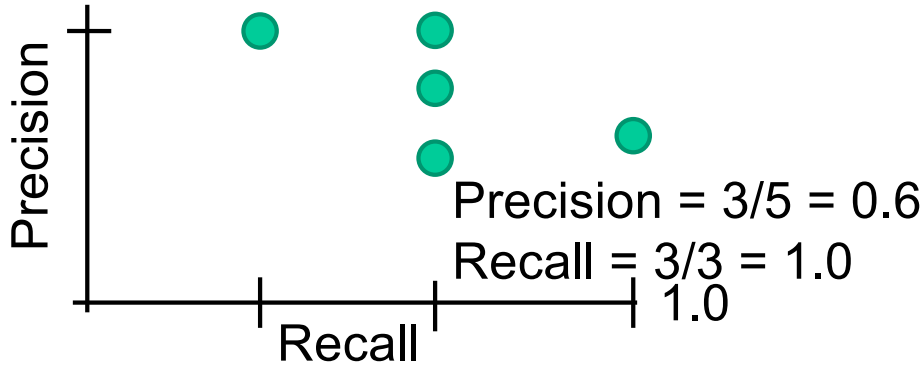


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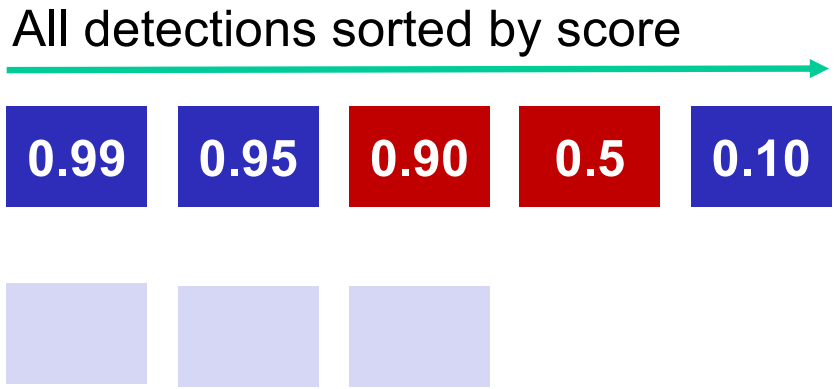
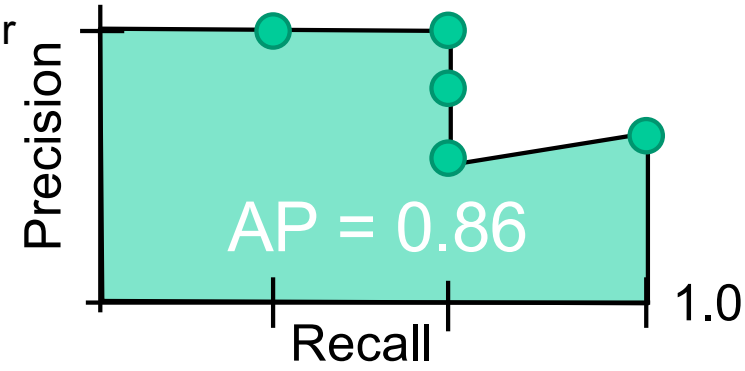
    2. Compute **Average Precision (AP)** or area under the Precision vs. Recall Curve**:**

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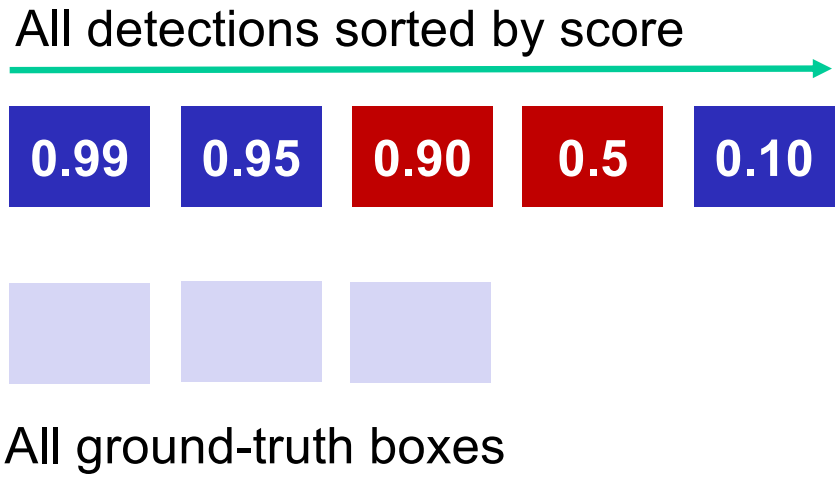
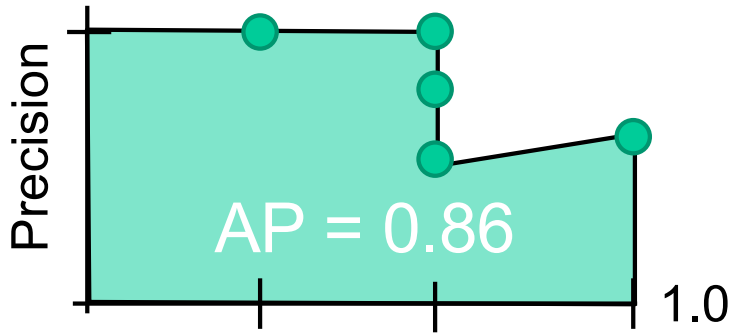



How to get AP = 1.0?

- Hit all GT boxes with IoU > 0.5, and have no “false positive” detections ranked above any “true positives”

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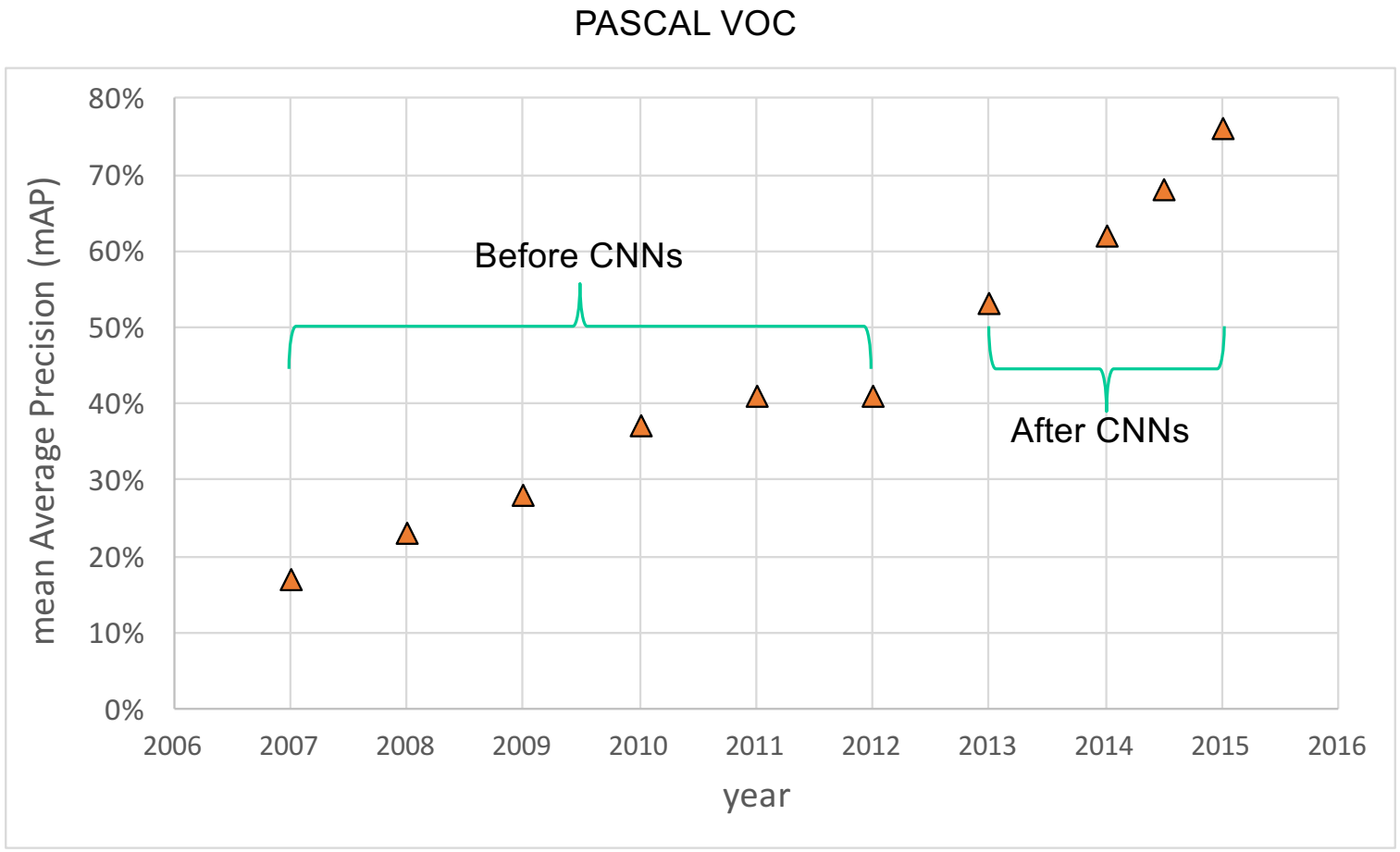
# PASCAL VOC Challenge (2005-2012)

- 20 challenge classes:
  - *Person*
  - *Animals:* bird, cat, cow, dog, horse, sheep
  - *Vehicles:* airplane, bicycle, boat, bus, car, motorbike, train
  - *Indoor:* bottle, chair, dining table, potted plant, sofa, tv/monitor

- Dataset size (by 2012): 11.5K training/validation images, 27K bounding boxes, 7K segmentations

http://host.robots.ox.ac.uk/pascal/VOC/

# Progress on PASCAL detection



# More recent benchmark: COCO

## What is COCO?

COCO is a large-scale object detection, segmentation, and captioning dataset. COCO has several features:

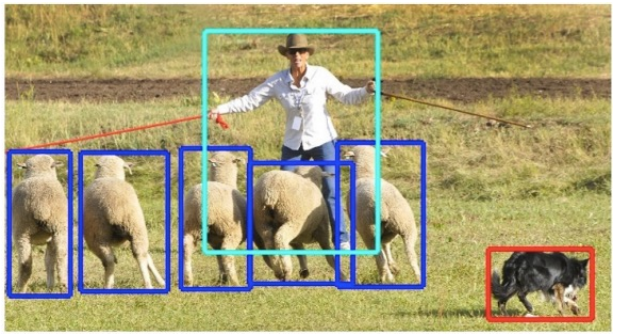
- **Object segmentation**
- **Recognition in context**
- **Superpixel stuff segmentation**
- **330K images (>200K labeled)**
- **1.5 million object instances**
- **80 object categories**
- **91 stuff categories**
- **5 captions per image**
- **250,000 people with keypoints**

COCO
Common Objects in Context


http://cocodataset.org/#home

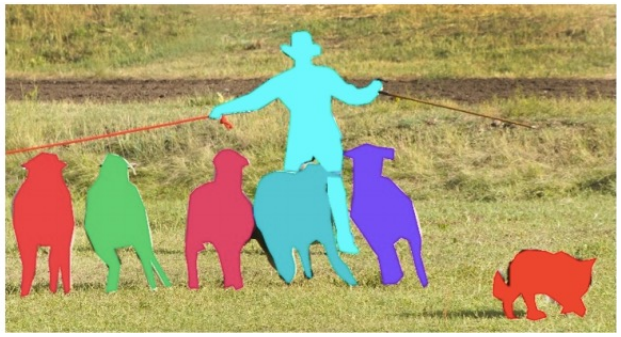
# COCO dataset: Tasks


image classification



object detection

semantic segmentation



instance segmentation

- Also: keypoint prediction, captioning, question answering…

# COCO detection metrics

```
Average Precision (AP):
  AP                % AP at IoU=.50:.05:.95 (primary challenge metric)
  AP^IoU=.50        % AP at IoU=.50 (PASCAL VOC metric)
  AP^IoU=.75        % AP at IoU=.75 (strict metric)
AP Across Scales:
  AP^small          % AP for small objects: area < 32^2
  AP^medium         % AP for medium objects: 32^2 < area < 96^2
  AP^large          % AP for large objects: area > 96^2
Average Recall (AR):
  AR^max=1          % AR given 1 detection per image
  AR^max=10         % AR given 10 detections per image
  AR^max=100        % AR given 100 detections per image
AR Across Scales:
  AR^small          % AR for small objects: area < 32^2
  AR^medium         % AR for medium objects: 32^2 < area < 96^2
  AR^large          % AR for large objects: area > 96^2
```

- Leaderboard: http://cocodataset.org/#detection-leaderboard
  - Not updated since 2020

# Object detection: Outline

- Task definition and evaluation
- Two-stage detectors: R-CNN family

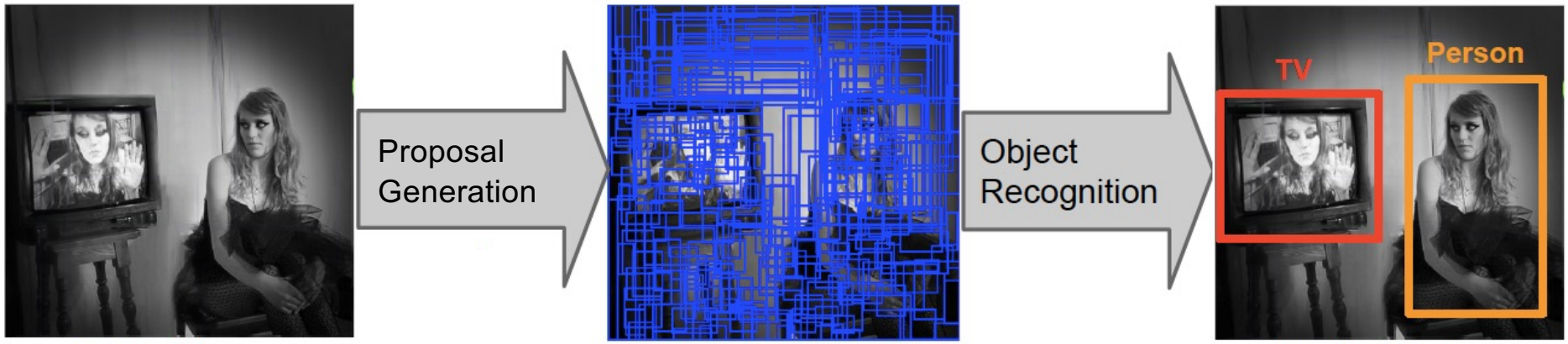
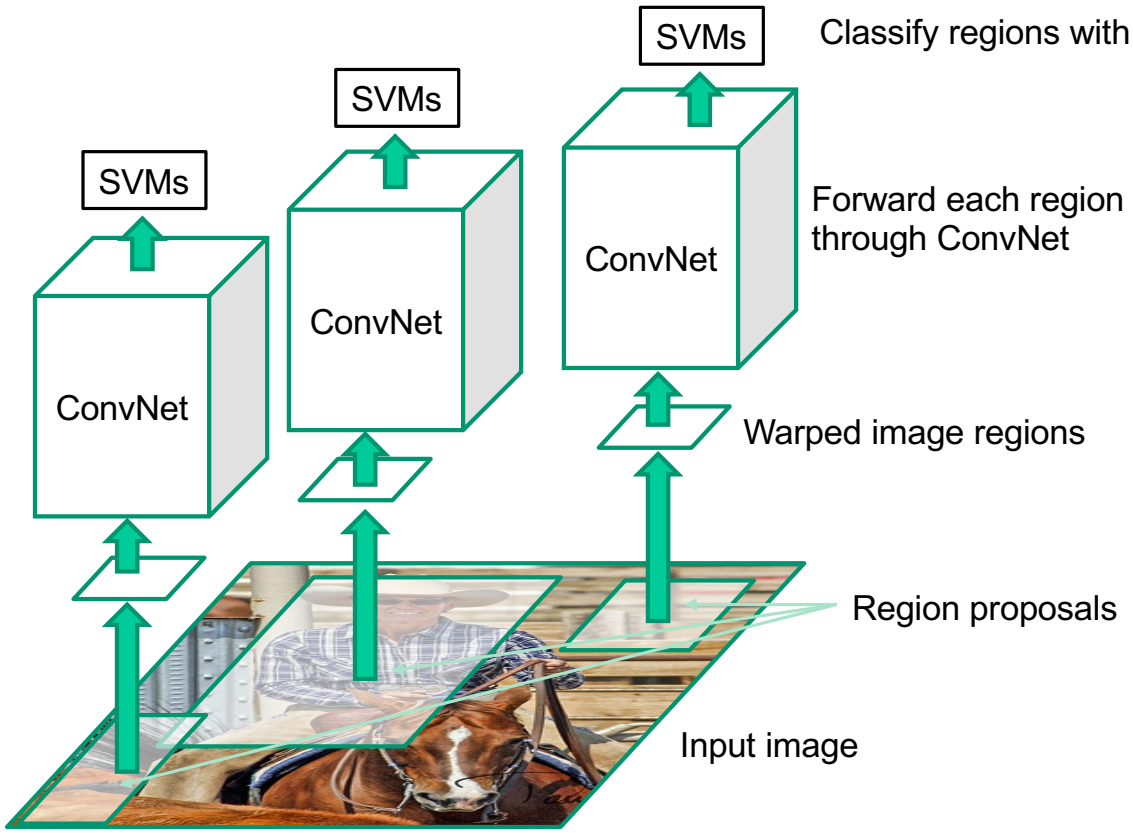



Image source

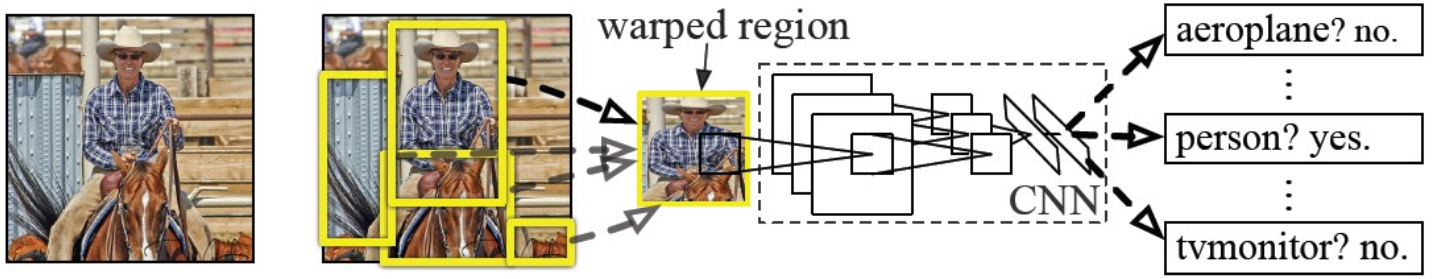
# R-CNN: Region proposals + CNN features

Source: R. Girshick




R. Girshick, J. Donahue, T. Darrell, and J. Malik, Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation, CVPR 2014

# R-CNN details

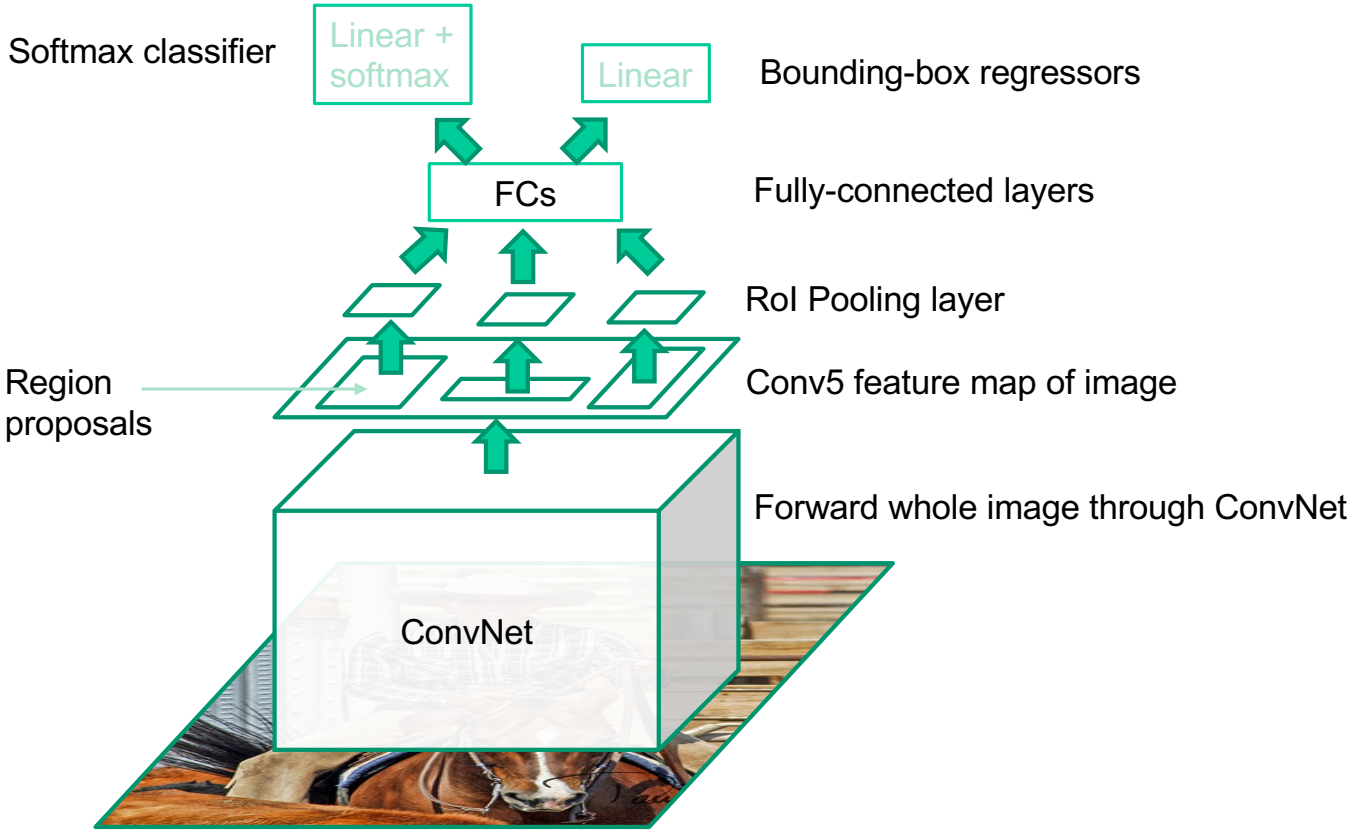



- **Regions**: ~2000 Selective Search proposals
- **Network**: AlexNet *pre-trained* on ImageNet (1000 classes), *fine-tuned* on PASCAL (21 classes)
- **Final detector**: warp proposal regions, extract fc7 network activations (4096 dimensions), classify with linear SVM
- **Bounding box regression** to refine box locations
- **Performance:** mAP of **53.7%** on PASCAL 2010
  (vs. **35.1%** for Selective Search and **33.4%** for Deformable Part Models)

# R-CNN pros and cons

- Pros
  - Much more accurate than previous approaches!
  - Any deep architecture can immediately be “plugged in”
- Cons
  - Not a single end-to-end system
    - Fine-tune network with softmax classifier (log loss)
    - Train post-hoc linear SVMs (hinge loss)
    - Train post-hoc bounding-box regressions (least squares)
  - Training was slow (84h), took up a lot of storage
    - 2000 CNN passes per image
  - Inference (detection) was slow (47s / image with VGG16)

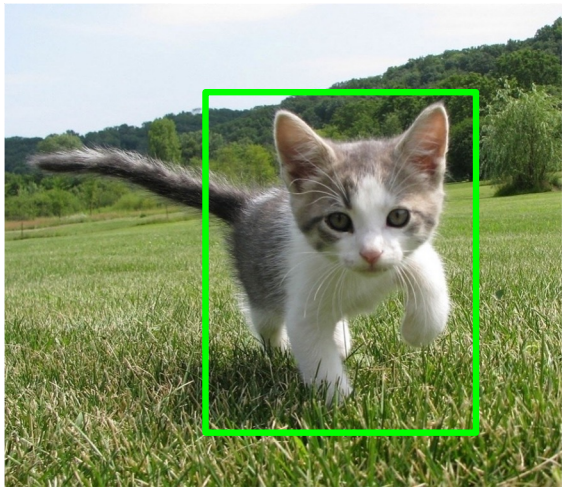
# Fast R-CNN




Source: R. Girshick

R. Girshick, Fast R-CNN, ICCV 2015

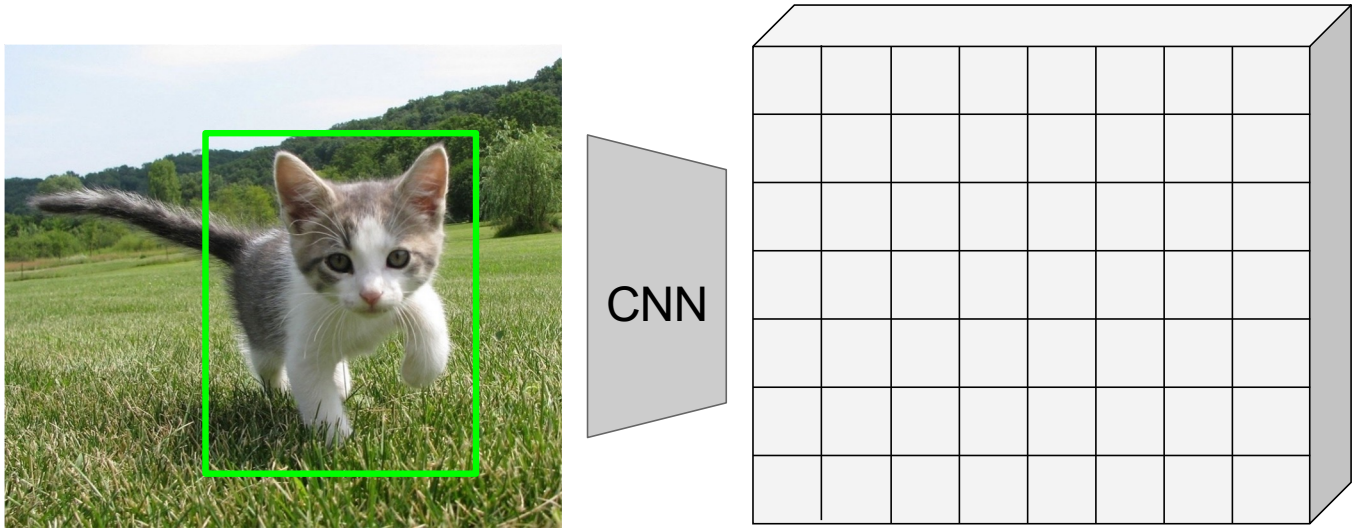
# RoI pooling



Input Image
(e.g., 3 x 640 x 480)

Source: J. Johnson

# RoI pooling

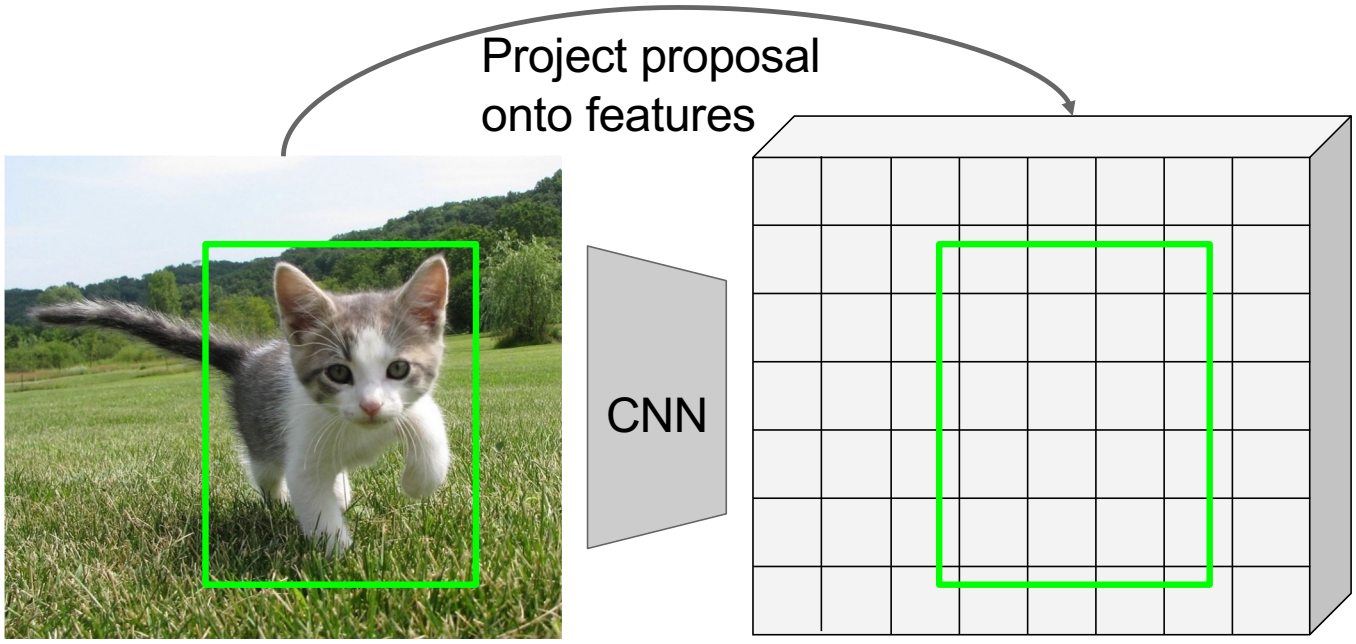



Input Image
(e.g., 3 x 640 x 480)

Feature map
(e.g., 512 x 20 x 15)

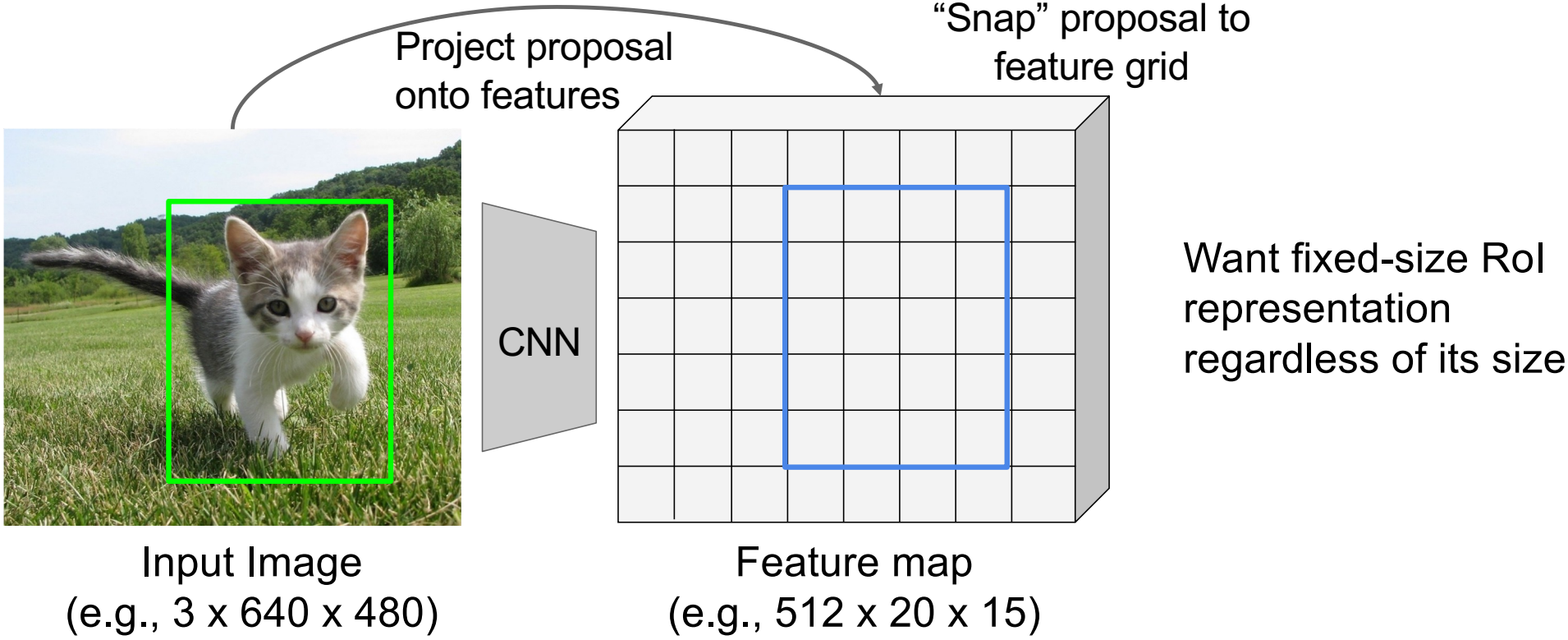
Source: J. Johnson

# RoI pooling




Source: J. Johnson

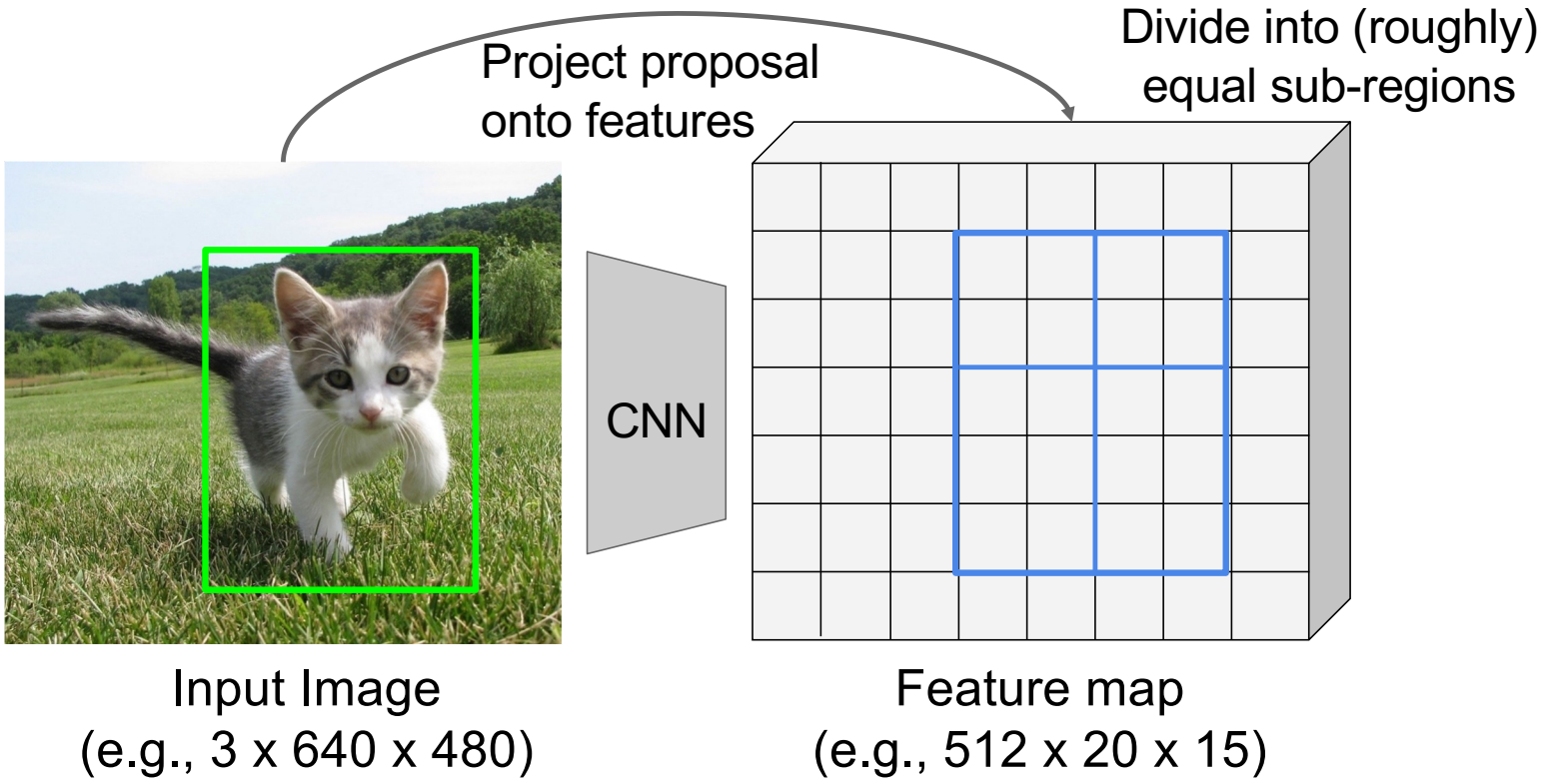
# RoI pooling




Want fixed-size RoI representation regardless of its size

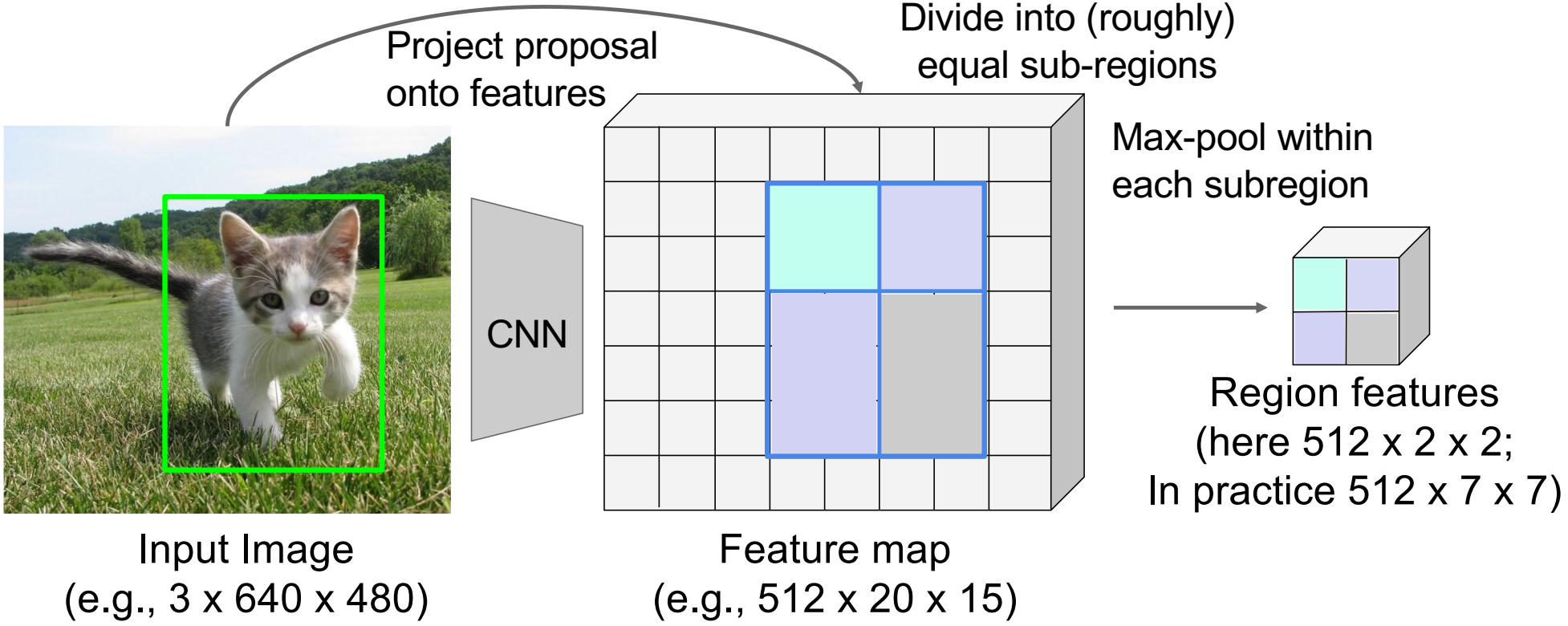
Source: J. Johnson

# RoI pooling




Source: J. Johnson

# RoI pooling




Source: J. Johnson

# RoI pooling illustration

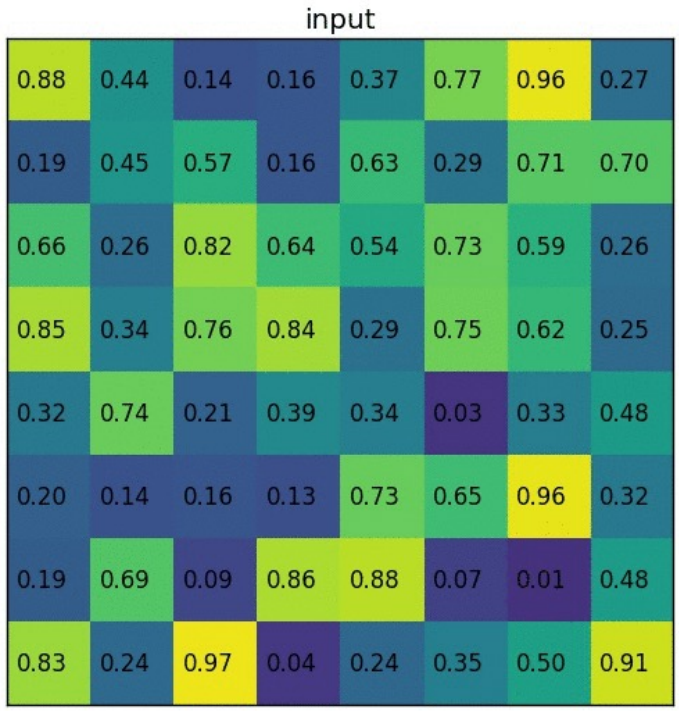
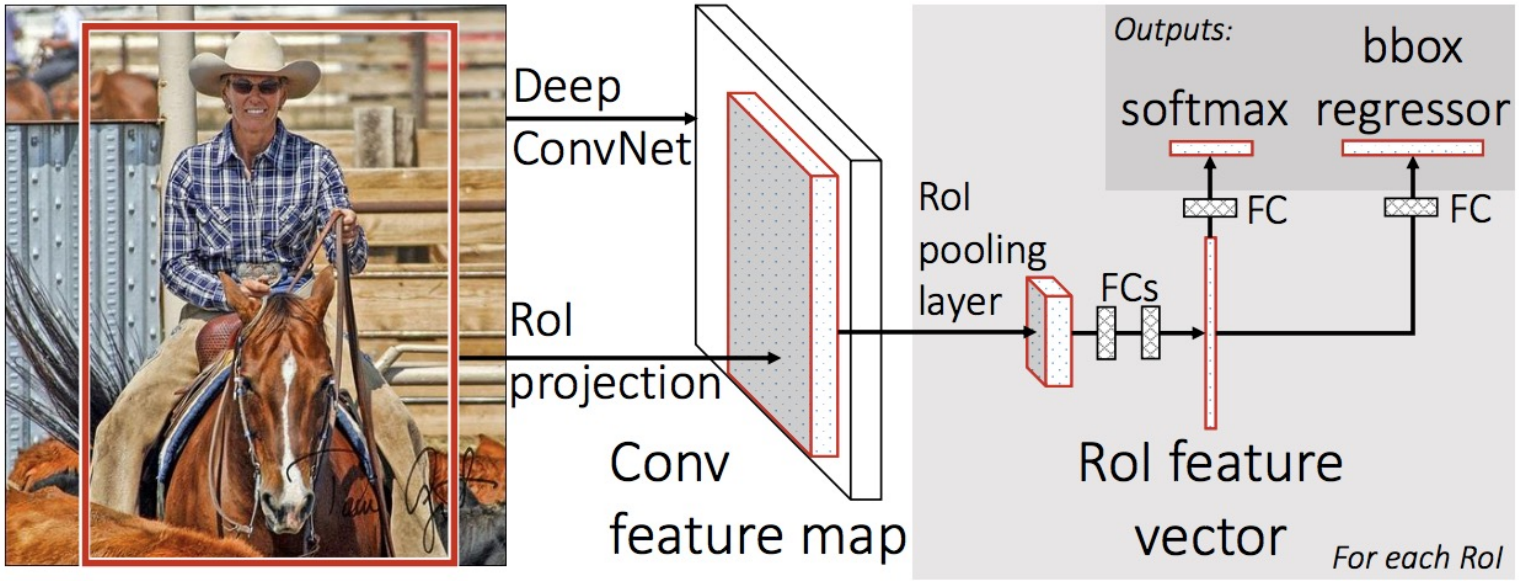



Image source

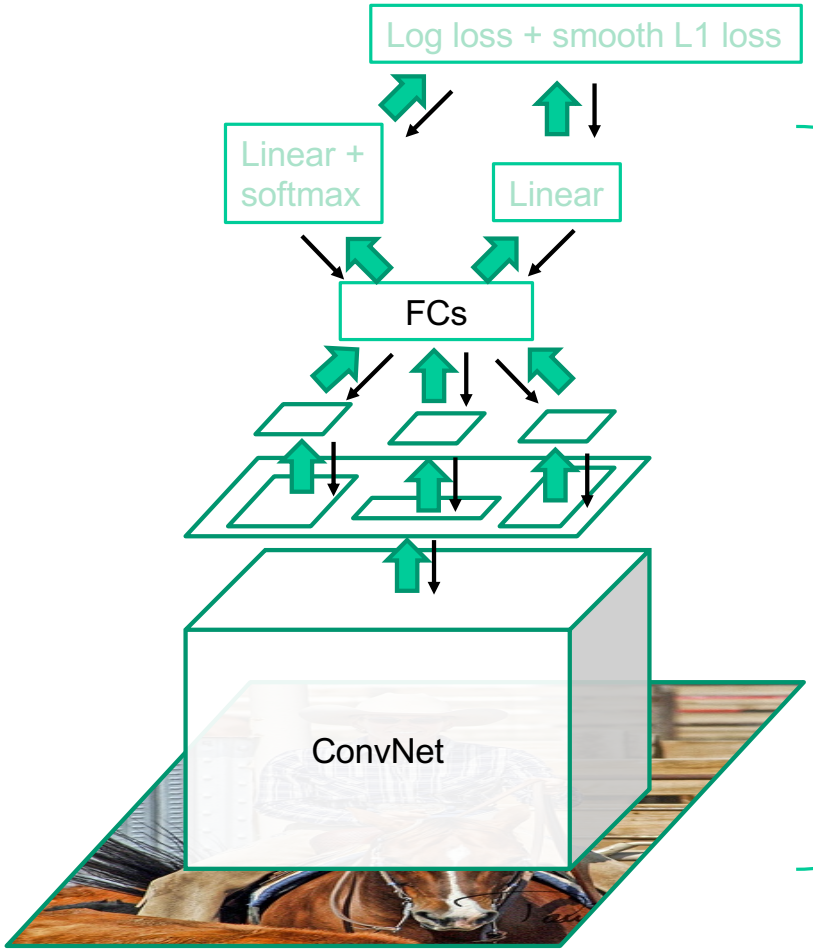
# Prediction

- For each RoI, network predicts probabilities for $C + 1$ classes (class 0 is background) and four bounding box offsets for $C$ classes




R. Girshick, Fast R-CNN, ICCV 2015

# Fast R-CNN training




*Multi-task* loss

Trainable

Source: R. Girshick

R. Girshick, Fast R-CNN, ICCV 2015

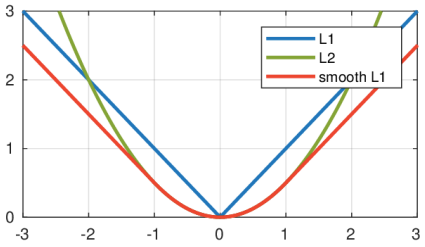
# Multi-task loss

- Loss for ground truth class $y$, predicted class probabilities $P(y)$, ground truth box $b$, and predicted box $\hat{b}$:

$$L(y, P, b, \hat{b}) = \underbrace{-\log P(y)}_{\text{softmax loss}} + \lambda \mathbb{I}[y \geq 1] \underbrace{L_{\text{reg}}(b, \hat{b})}_{\text{regression loss}}$$

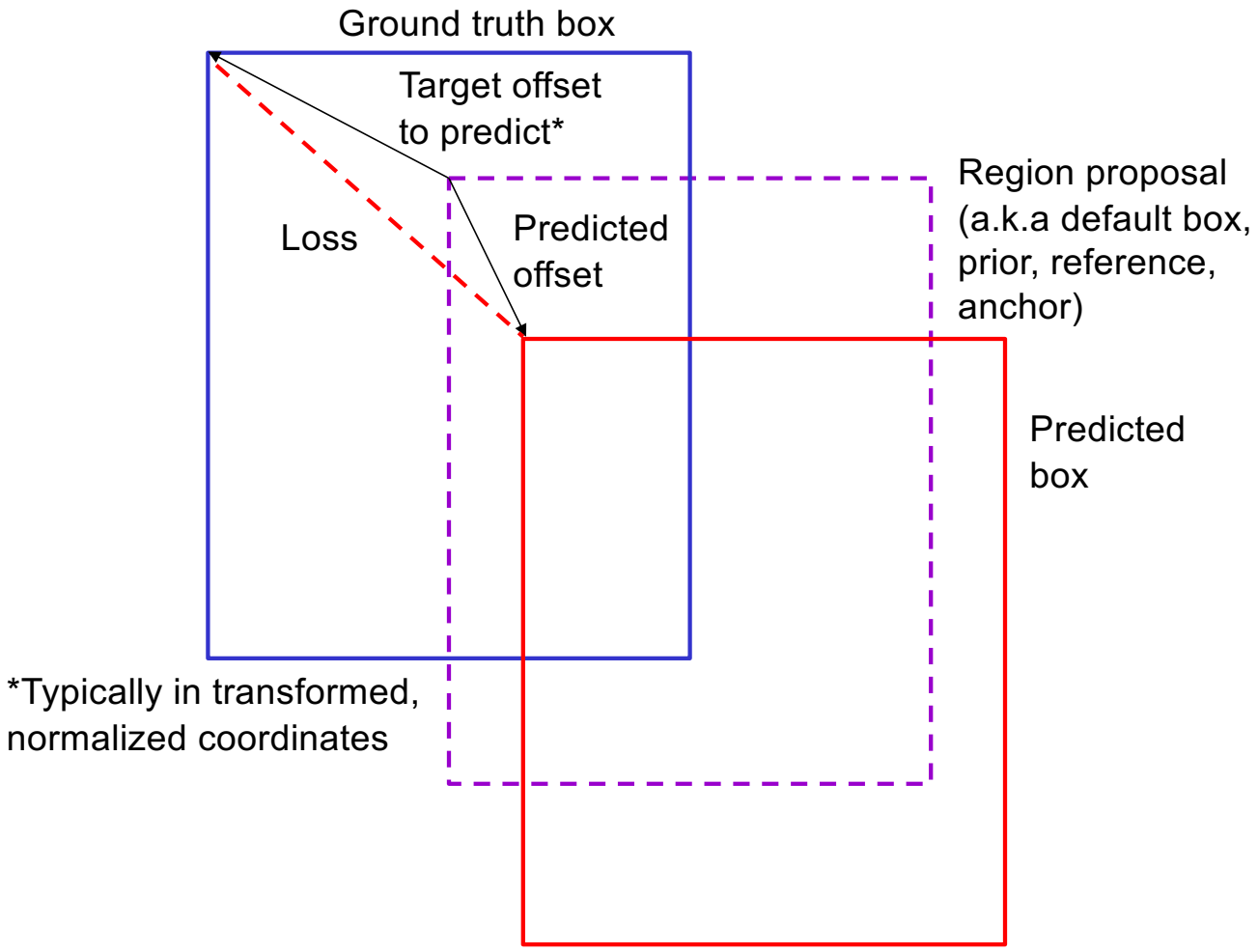
- Regression loss: *smooth* $L_1$ *loss* on top of log space offsets relative to proposal

$$L_{\text{reg}}(b, \hat{b}) = \sum_{i=\{x,y,w,h\}} \text{smooth}_{L_1}(b_i - \hat{b}_i)$$




$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1 \\ |x| - 0.5 & \text{otherwise} \end{cases}$$

# Bounding box regression



# Fast R-CNN results

| | Fast R-CNN | R-CNN |
|---|---|---|
| Train time (h) | **9.5** | 84 |
| - Speedup | **8.8x** | |
| Test time / image | **0.32s** | 47.0s |
| - Test speedup | **146x** | |
| mAP | **66.9%** | 66.0% |

(vs. 53.7% for AlexNet)

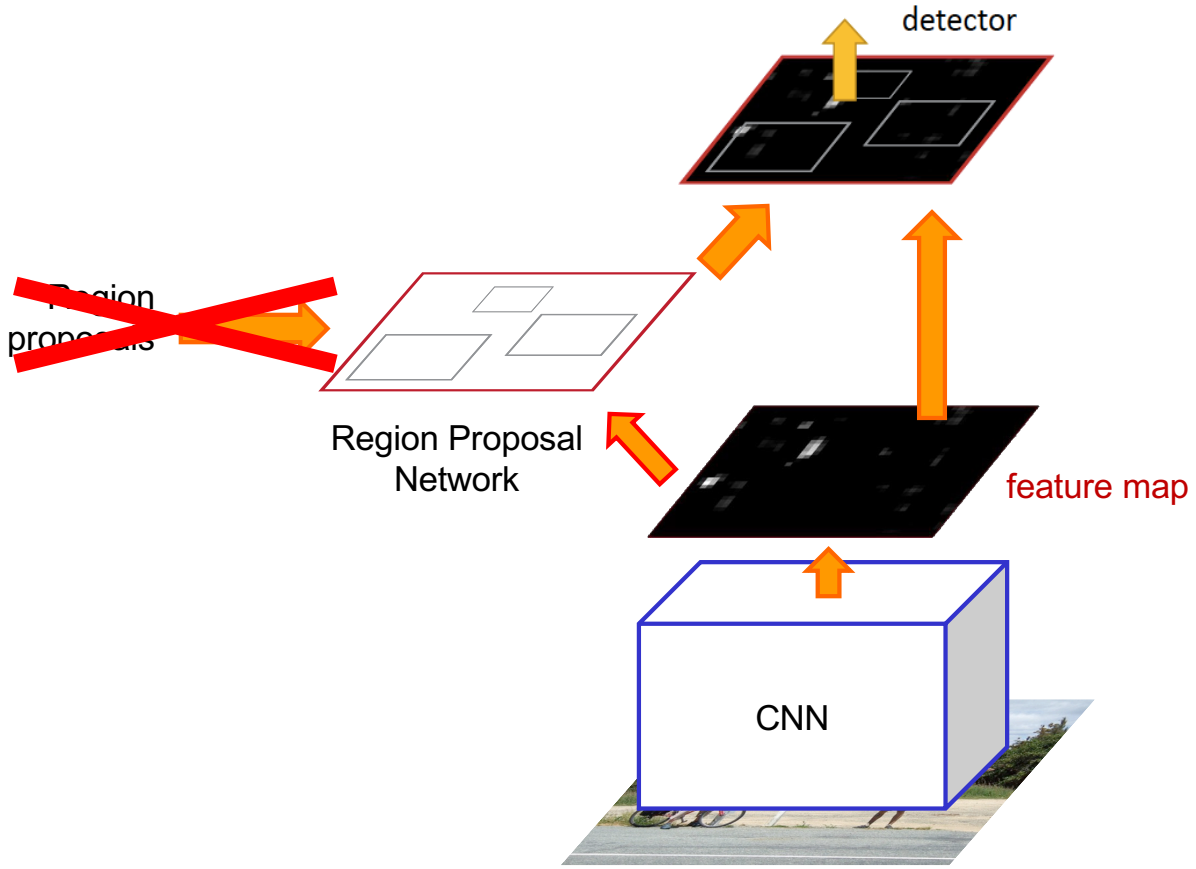
Timings exclude object proposal time, which is equal for all methods.
All methods use VGG16.

Source: R. Girshick, K. He

# Object detection: Outline

- Task definition and evaluation
- Two-stage detectors:
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN

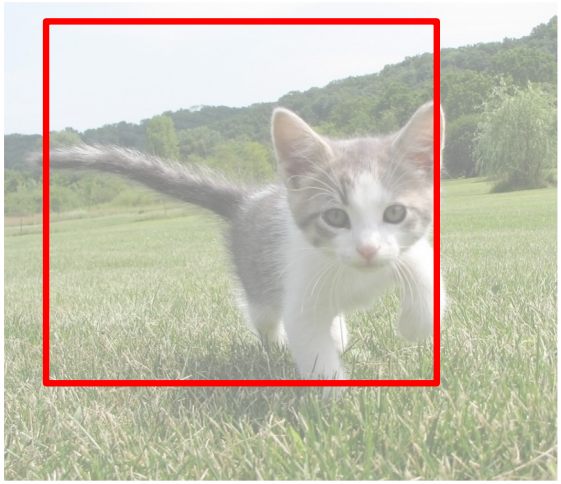
# Faster R-CNN




S. Ren, K. He, R. Girshick, and J. Sun, Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks, NIPS 2015

# Region proposal network (RPN)

- Idea: tile the image with “anchor boxes” of a set size and try to predict how likely each anchor is to contain an object



**Anchor is an object?**

Figure source: J. Johnson

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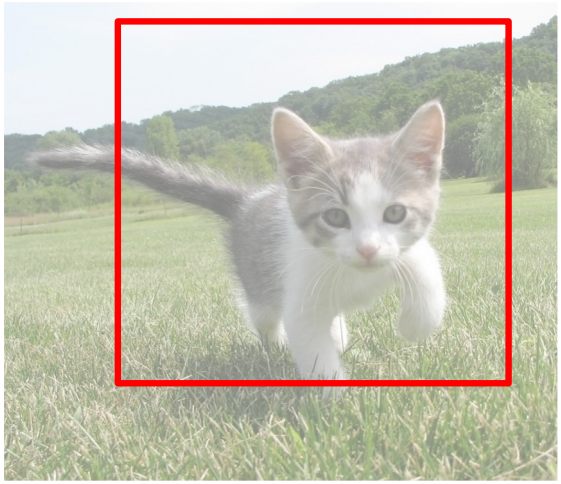

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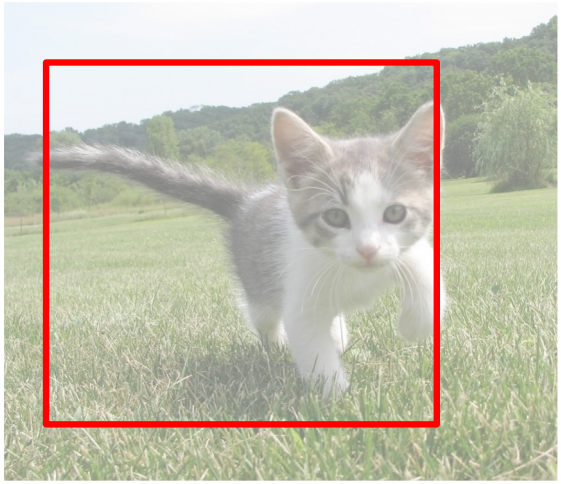

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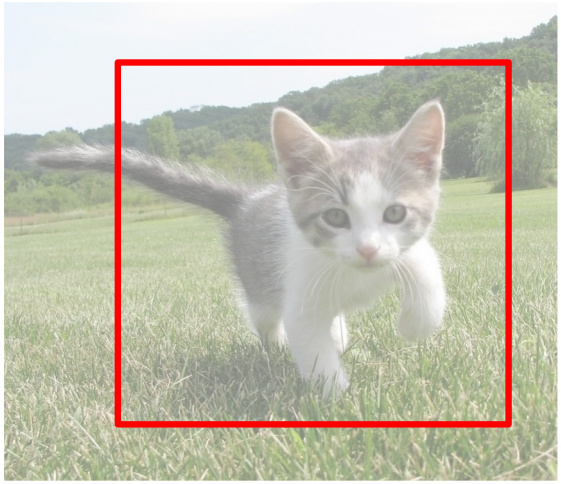
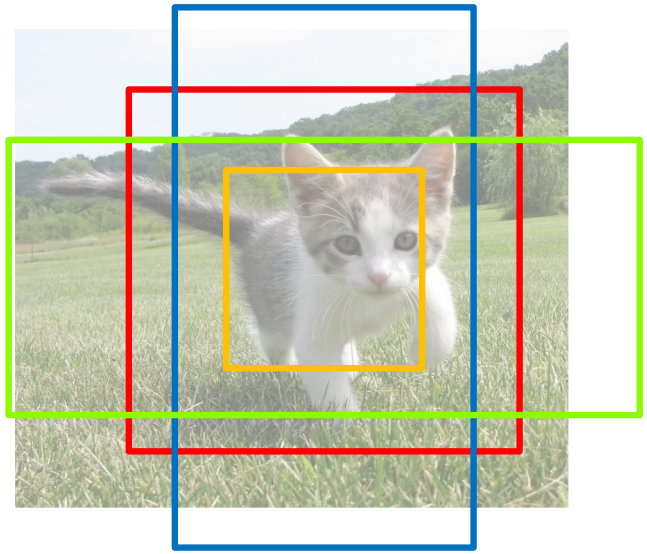


**Anchor is an object?**

Figure source: J. Johnson

# Region proposal network (RPN)

- Idea: tile the image with “anchor boxes” of a set size and try to predict how likely each anchor is to contain an object
- Introduce anchor boxes at multiple scales and aspect ratios to handle a wider range of object sizes and shapes

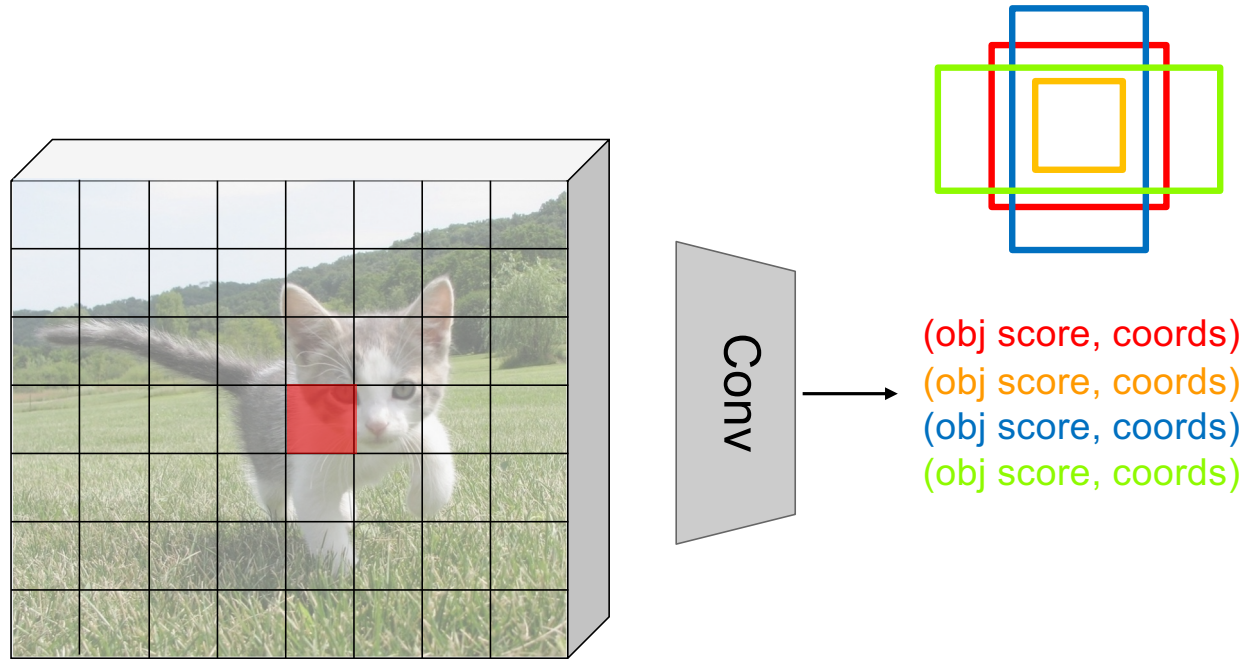


**Anchor is an object?**
**Anchor is an object?**
**Anchor is an object?**
**Anchor is an object?**

Figure source: J. Johnson

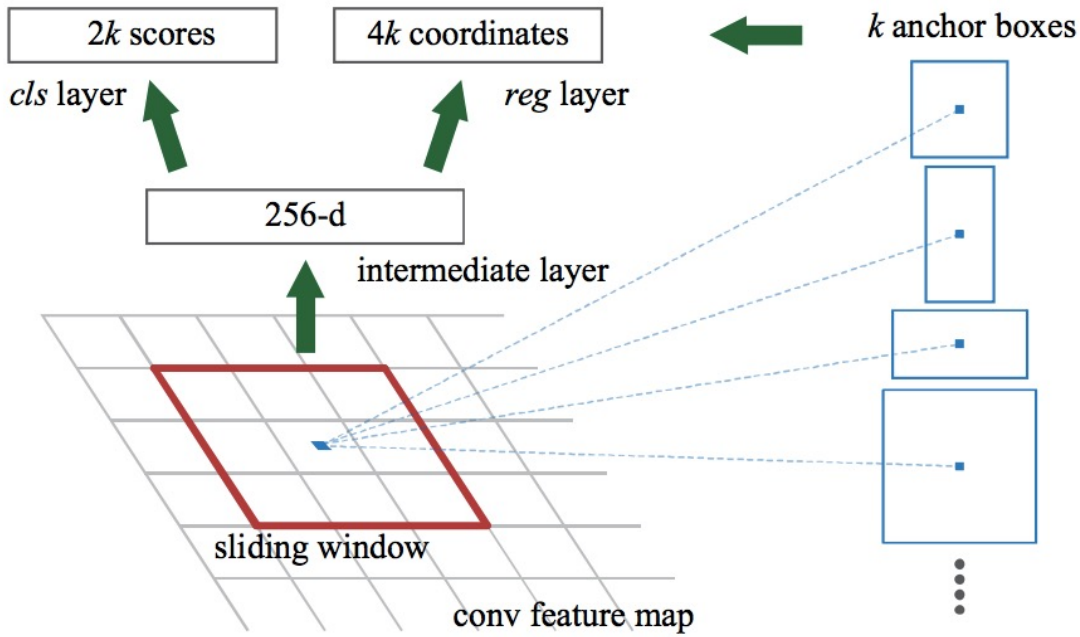
# Region proposal network (RPN)

- Implementation: put conv layers over low-resolution feature grid, for each grid location predict “object/no object” scores and bounding box regression coordinates

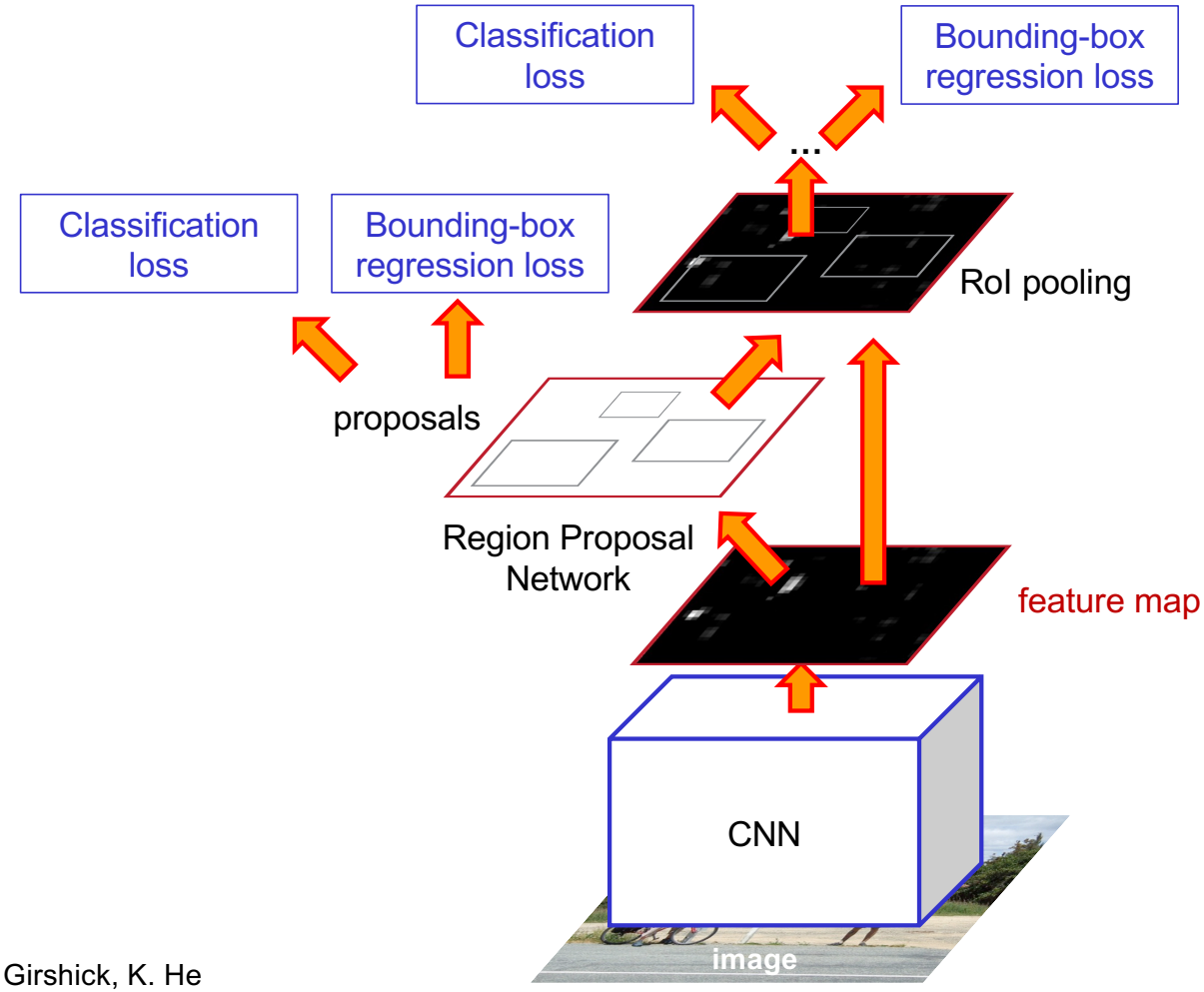


# Faster R-CNN RPN design

- Slide a small window (3x3) over the conv5 layer
  - Predict object/no object
  - Regress bounding box coordinates with reference to *anchors* (3 scales x 3 aspect ratios)



# One network, four losses



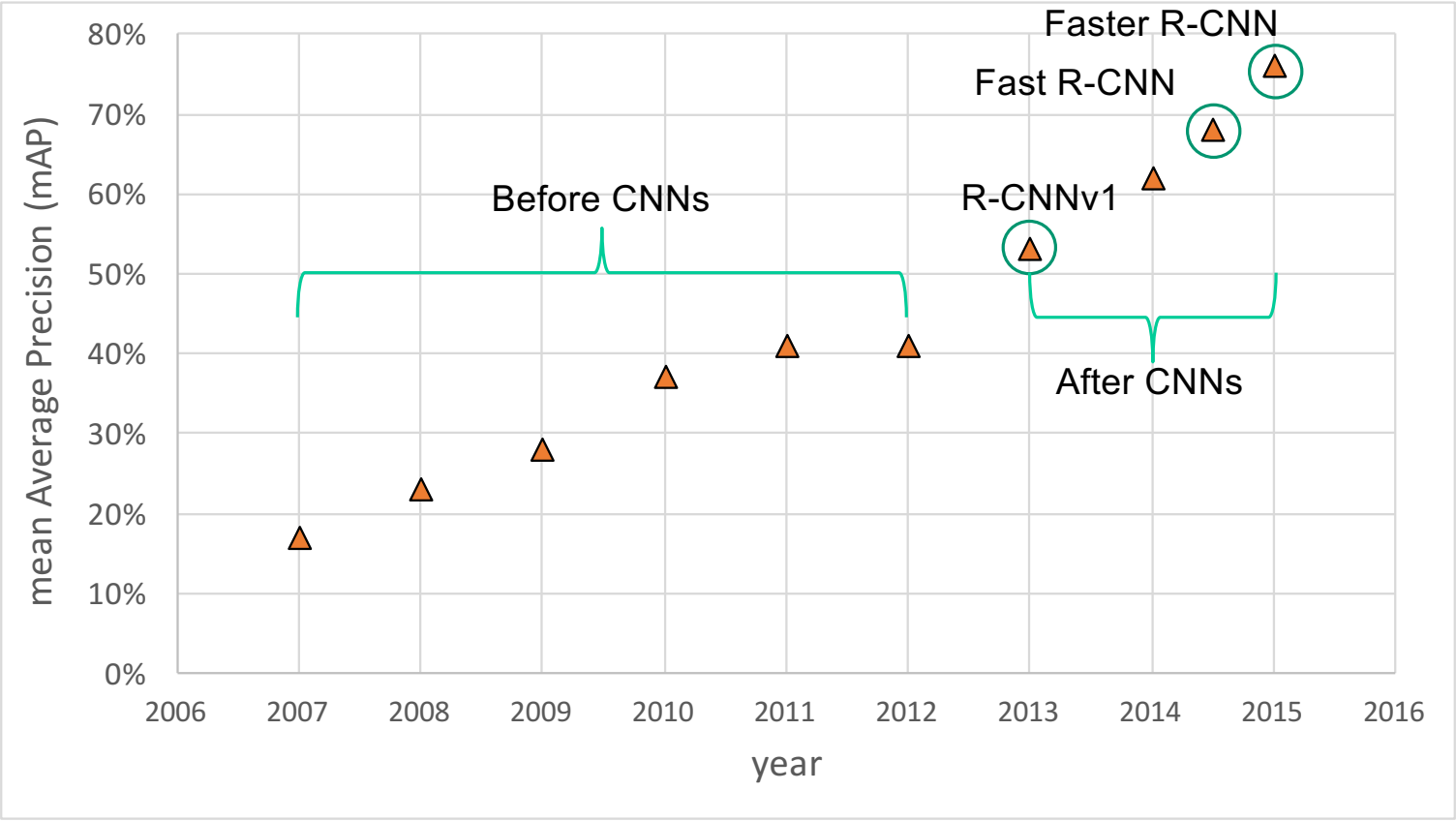

Source: R. Girshick, K. He

# Faster R-CNN results

| system | time | 07 data | 07+12 data |
|---|---|---|---|
| R-CNN | ~50s | 66.0 | - |
| Fast R-CNN | ~2s | 66.9 | 70.0 |
| Faster R-CNN | 198ms | **69.9** | **73.2** |

detection mAP on PASCAL VOC 2007, with VGG-16 pre-trained on ImageNet

# Object detection progress



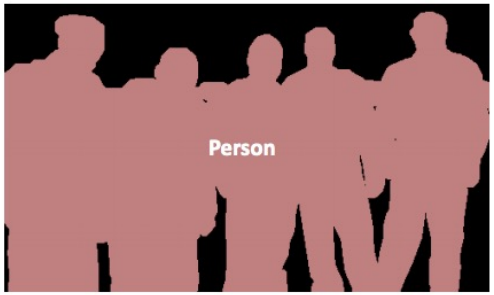
# Object detection: Outline

- Task definition and evaluation
- Two-stage detectors:
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - Mask R-CNN

# Instance segmentation

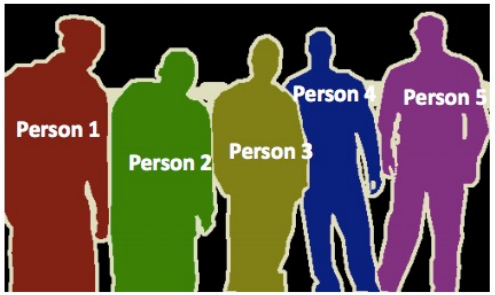
Object Detection

✓




Semantic Segmentation

✓



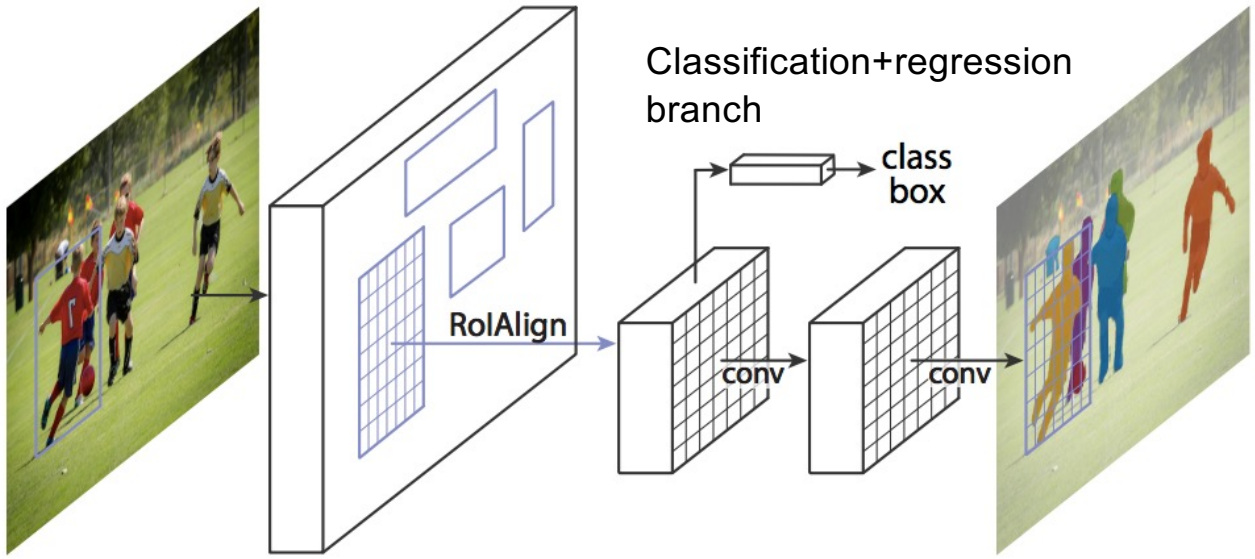

**Instance Segmentation**

?

Source: Kaiming He

# Mask R-CNN

- Mask R-CNN = Faster R-CNN + FCN on RoIs

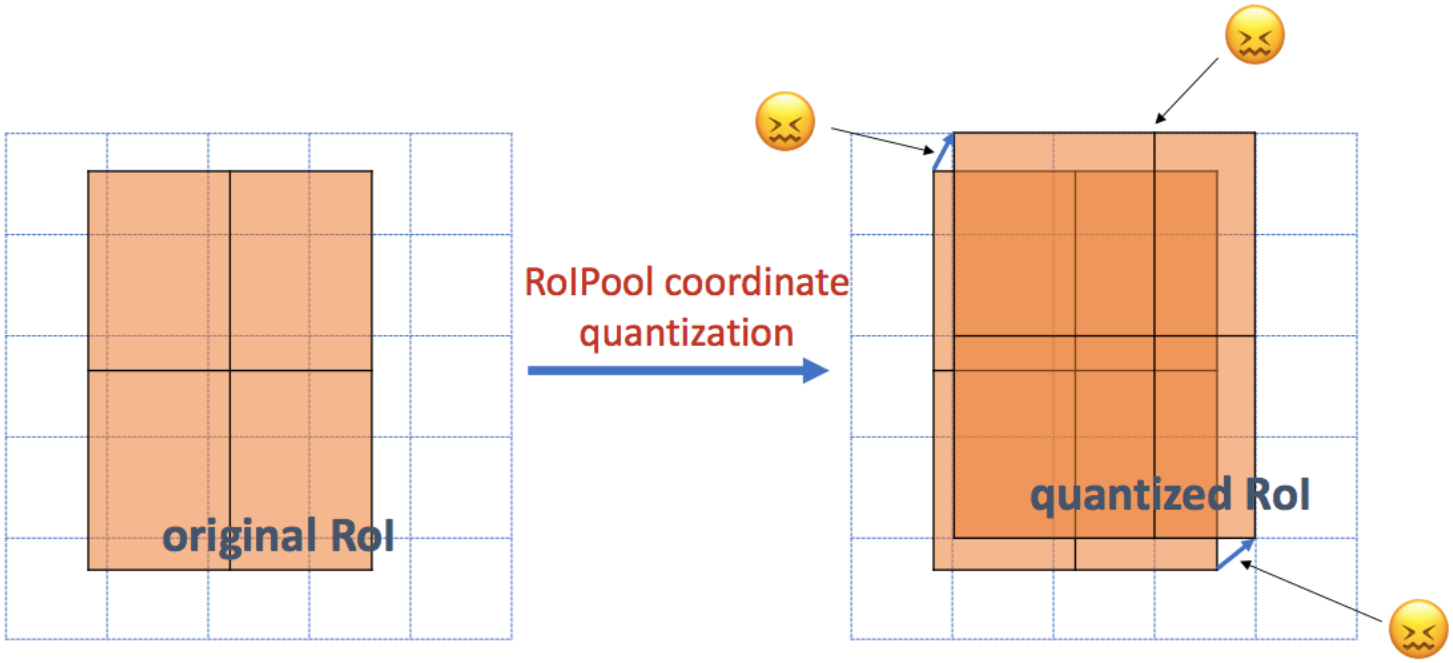



Mask branch: separately predict segmentation for each possible class

K. He, G. Gkioxari, P. Dollar, and R. Girshick, Mask R-CNN,
ICCV 2017 (Best Paper Award)

# RoIAlign vs. RoIPool

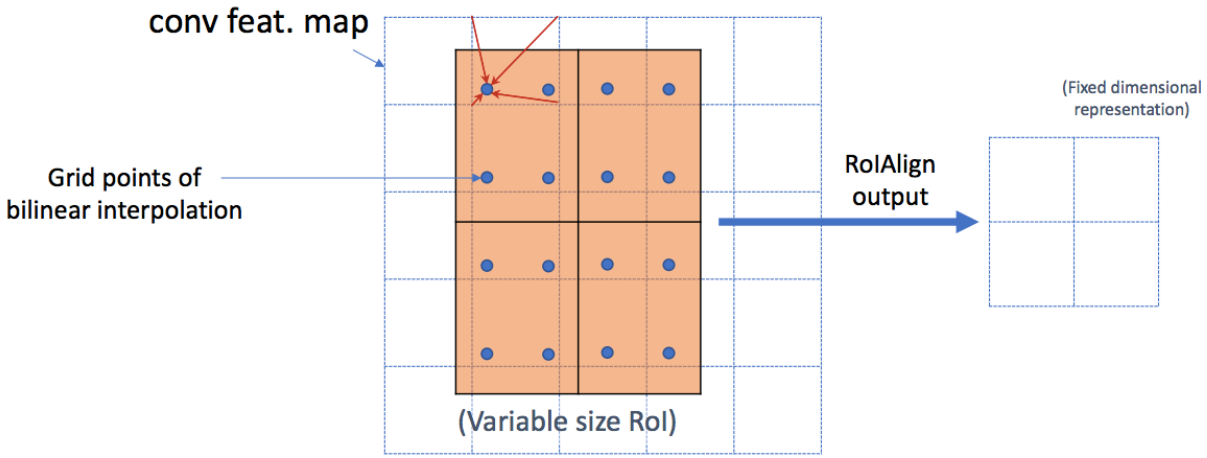
- RoIPool: nearest neighbor quantization




Source: K. He, R. Girshick

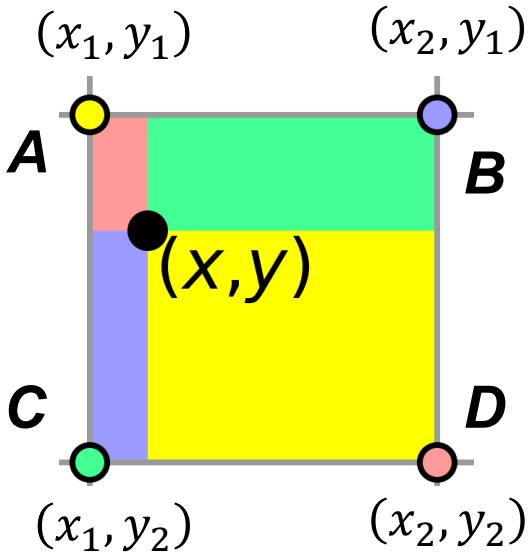
# RoIAlign vs. RoIPool

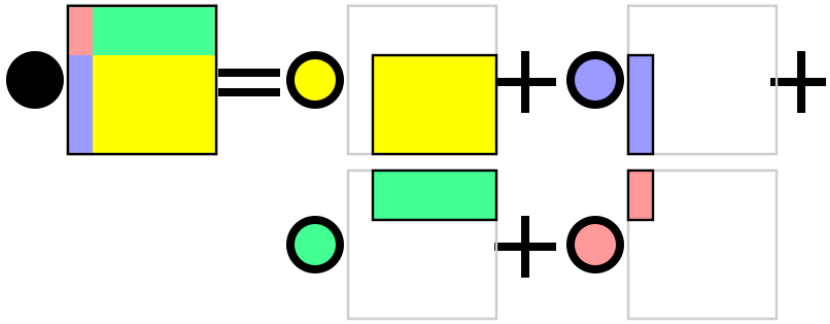
- RoIPool: nearest neighbor quantization
- RoIAlign: bilinear interpolation




Source: K. He, R. Girshick

# Bilinear interpolation



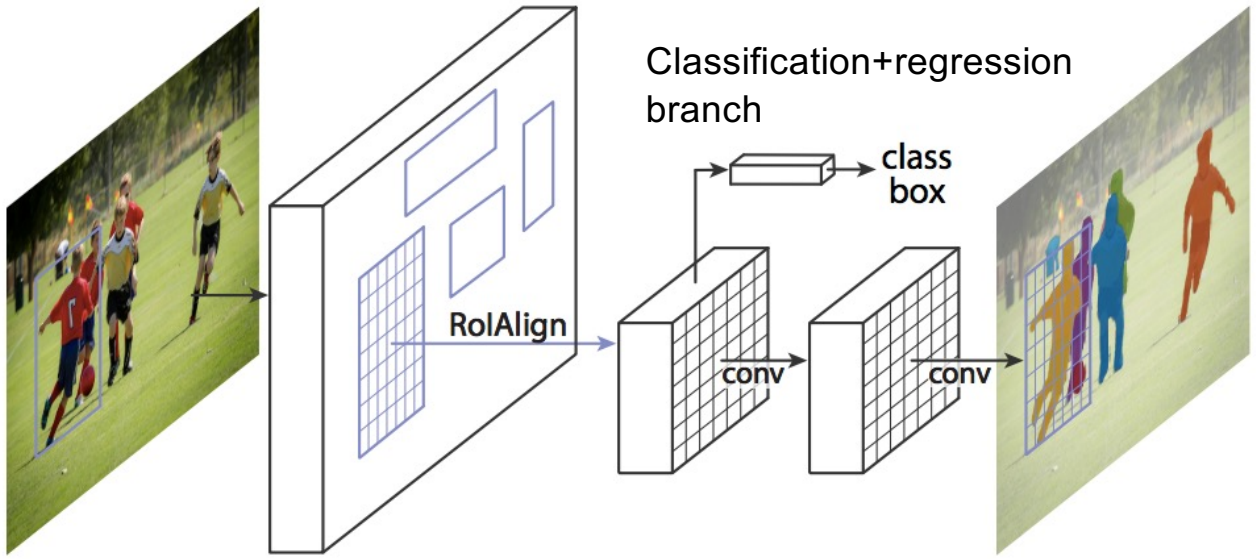
$$f(x,y) = w_{11}A + w_{21}B + w_{12}C + w_{22}D$$

$$w_{11} = \frac{(x_2 - x)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)} \qquad w_{21} = \frac{(x - x_1)(y_2 - y)}{(x_2 - x_1)(y_2 - y_1)}$$

$$w_{12} = \frac{(x_2 - x)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)} \qquad w_{22} = \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}$$

http://en.wikipedia.org/wiki/Bilinear_interpolation

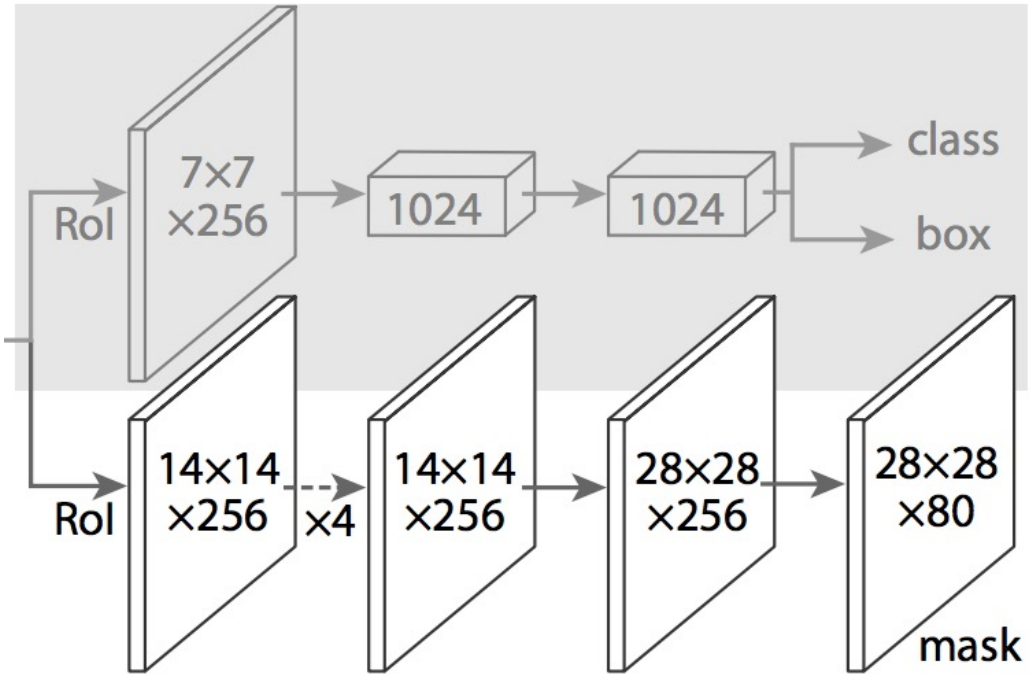
# Mask R-CNN




Mask branch: separately predict segmentation for each possible class

# Mask R-CNN

- From RoIAlign features, predict class label, bounding box, and segmentation mask



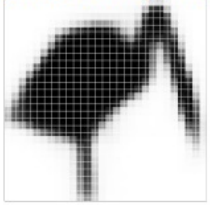

Classification/regression head from an established object detector (e.g., FPN)

Separately predict binary mask for each class with per-pixel sigmoids, use average binary cross-entropy loss

# Mask R-CNN

Validation image with box detection shown in red

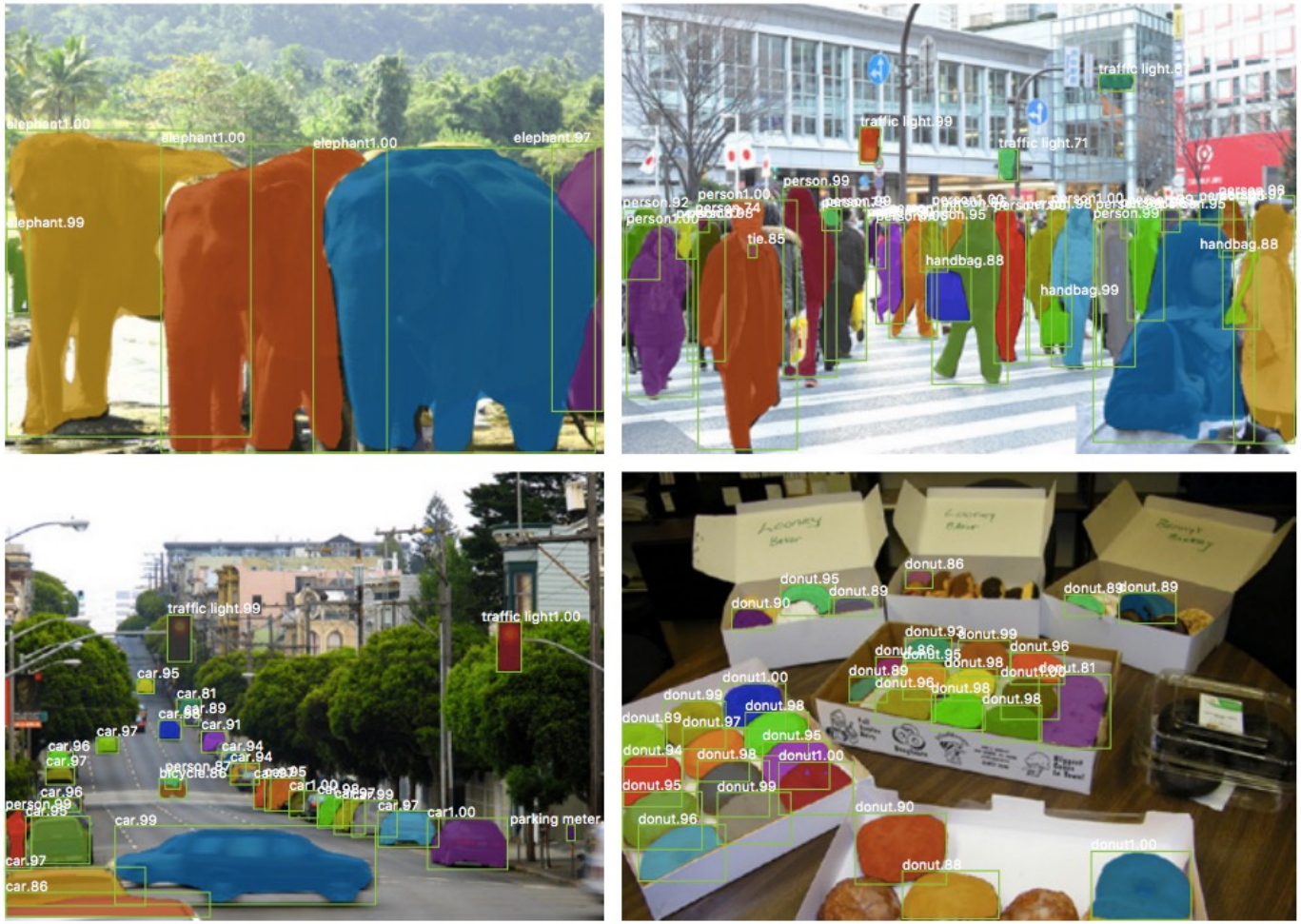
28x28 soft prediction



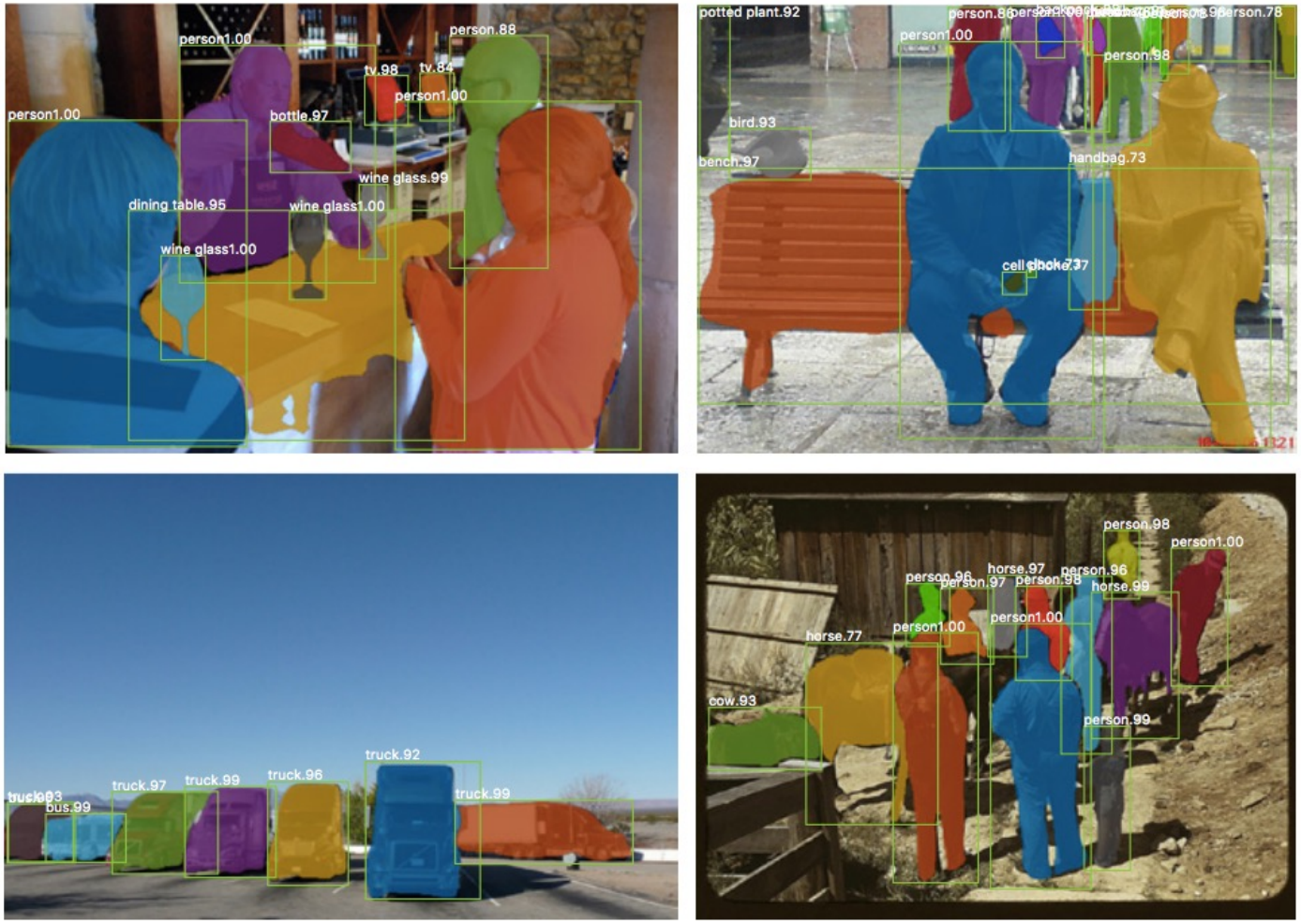
Resized Soft prediction

Final mask

# Example results



# Example results



# Instance segmentation results on COCO

| | backbone | AP | $AP_{50}$ | $AP_{75}$ | $AP_S$ | $AP_M$ | $AP_L$ |
|---|---|---|---|---|---|---|---|
| MNC [10] | ResNet-101-C4 | 24.6 | 44.3 | 24.8 | 4.7 | 25.9 | 43.6 |
| FCIS [26] +OHEM | ResNet-101-C5-dilated | 29.2 | 49.5 | - | 7.1 | 31.3 | 50.0 |
| FCIS+++ [26] +OHEM | ResNet-101-C5-dilated | 33.6 | 54.5 | - | - | - | - |
| **Mask R-CNN** | ResNet-101-C4 | 33.1 | 54.9 | 34.8 | 12.1 | 35.6 | 51.1 |
| **Mask R-CNN** | ResNet-101-FPN | 35.7 | 58.0 | 37.8 | 15.5 | 38.1 | 52.4 |
| **Mask R-CNN** | ResNeXt-101-FPN | **37.1** | **60.0** | **39.4** | **16.9** | **39.9** | **53.5** |

AP at different IoU thresholds

AP for different size instances

# Keypoint prediction

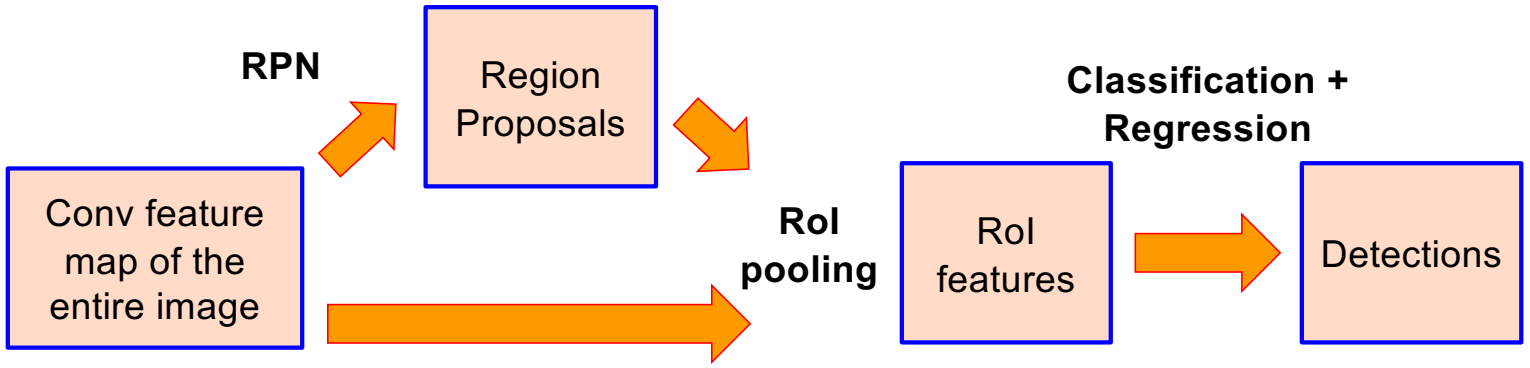
- Given $K$ keypoints, train model to predict $K$ $m \times m$ *one-hot* maps with cross-entropy losses over $m^2$ outputs

# Outline

- Task definition and evaluation
- Two-stage detectors
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - Mask R-CNN
- Single-stage and multi-resolution detectors

# Streamlined detection architectures

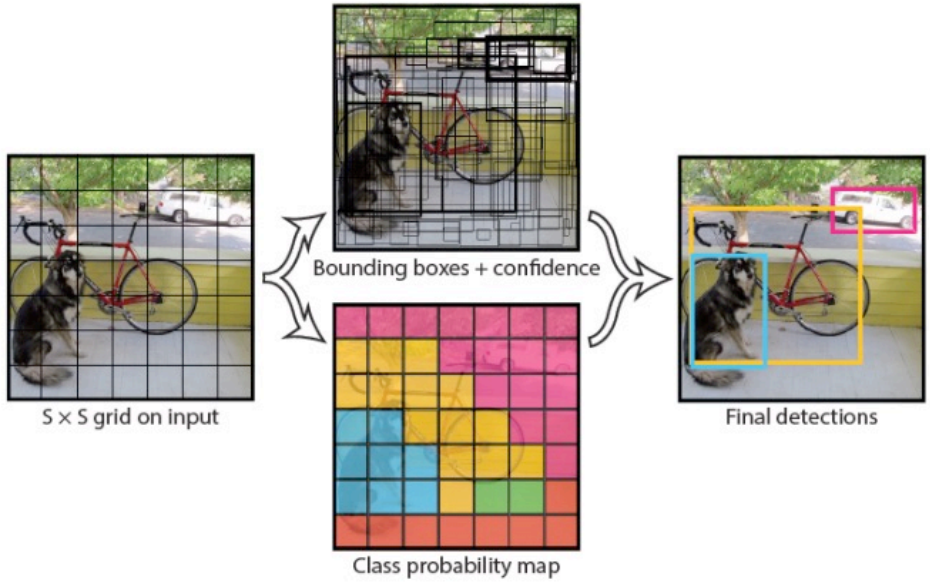
- The Faster R-CNN pipeline separates proposal generation and region classification




- Is it possible to do detection in one shot?

# YOLO

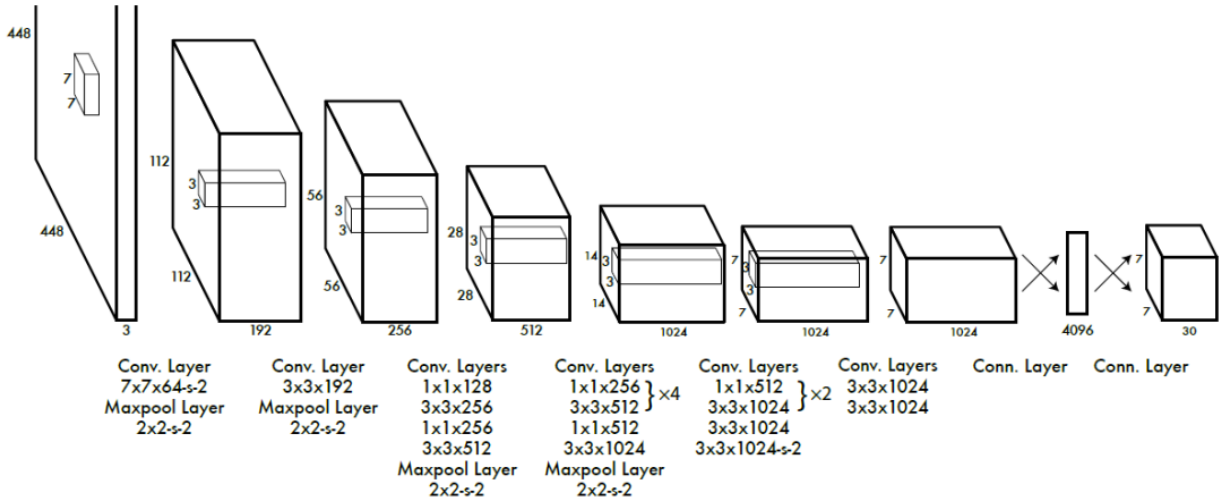
- Divide the image into a coarse grid and directly predict class label and a few candidate boxes for each grid cell




J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, You Only Look Once: Unified, Real-Time Object Detection, CVPR 2016

# YOLO

1. Take conv feature maps at 7x7 resolution
2. Add two FC layers to predict, at each location, a score for each class and 2 bboxes w/ confidences
   - For PASCAL, output is $7\times7\times30$ $(30 = 20 + 2 * (4 + 1))$




J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, You Only Look Once: Unified, Real-Time Object Detection, CVPR 2016

# YOLO

- Objective function:

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[ (x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[ \left( \sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left( \sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Regression

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left( C_i - \hat{C}_i \right)^2$$

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left( C_i - \hat{C}_i \right)^2$$

Object/no object confidence

$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} \left( p_i(c) - \hat{p}_i(c) \right)^2$$

Class prediction

# YOLO

- Objective function:

Cell i contains object, predictor j is responsible for it

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[ (x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[ \left( \sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left( \sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Small deviations matter less for larger boxes than for smaller boxes

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left( C_i - \hat{C}_i \right)^2$$

Confidence for object

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left( C_i - \hat{C}_i \right)^2$$

Confidence for no object

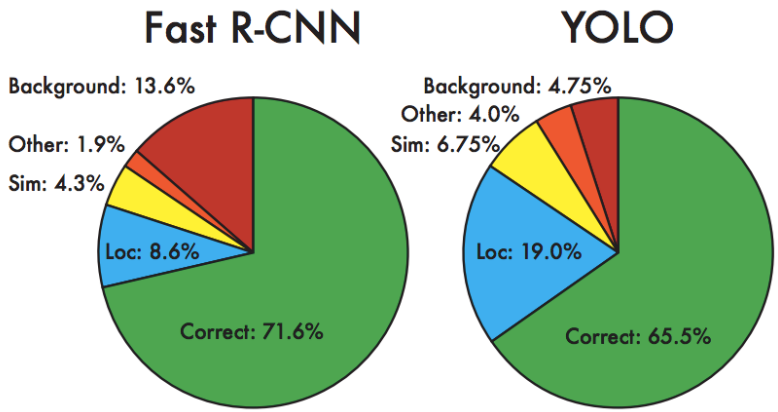
Down-weight loss from boxes that don't contain objects ($\lambda_{\text{noobj}} = 0.5$)

$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} \left( p_i(c) - \hat{p}_i(c) \right)^2$$

Class probability

# YOLO: Results

- Each grid cell predicts only two boxes and can only have one class – this limits the number of nearby objects that can be predicted
- Localization accuracy suffers compared to Fast(er) R-CNN due to coarser features, errors on small boxes
- 7x speedup over Faster R-CNN (45-155 FPS vs. 7-18 FPS)

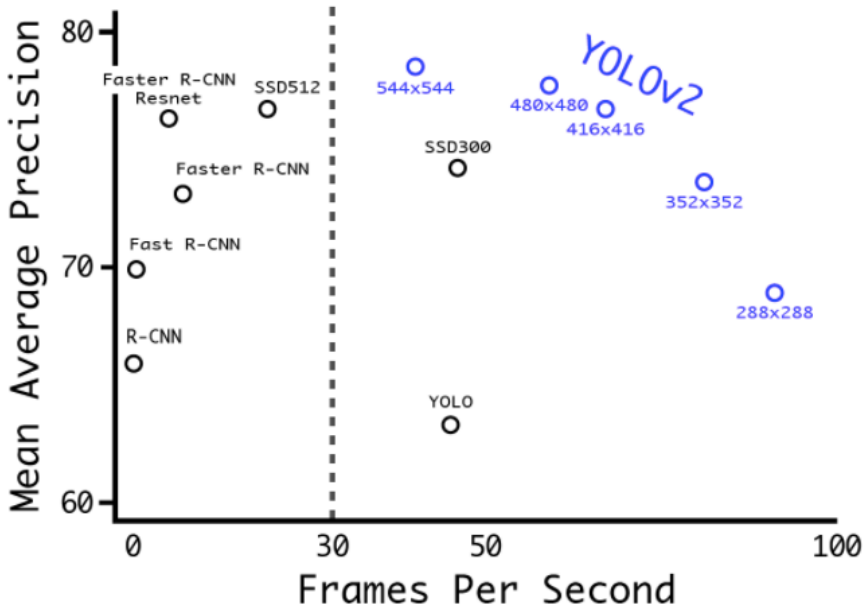



**Performance on PASCAL 2007**

# YOLO v2

- Remove FC layer, do convolutional prediction with anchor boxes instead
- Increase resolution of input images and conv feature maps
- Improve accuracy using batch normalization and other tricks

**VOC 2007 results**

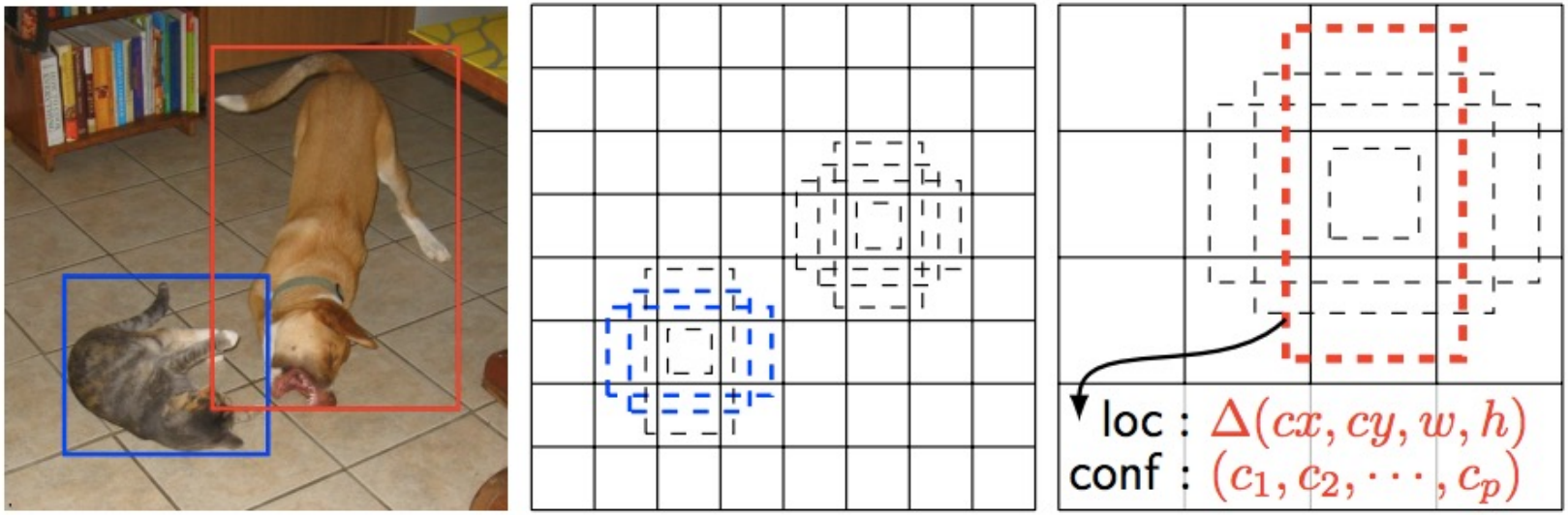



**YouTube demo**

J. Redmon and A. Farhadi, YOLO9000: Better, Faster, Stronger, CVPR 2017

# Multi-resolution prediction: SSD

- Predict boxes of different size from different conv maps
- Each level of resolution has its own predictor




(a) Image with GT boxes (b) $8 \times 8$ feature map (c) $4 \times 4$ feature map

W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. Berg, SSD: Single Shot MultiBox Detector, ECCV 2016

# Multi-resolution prediction: SSD

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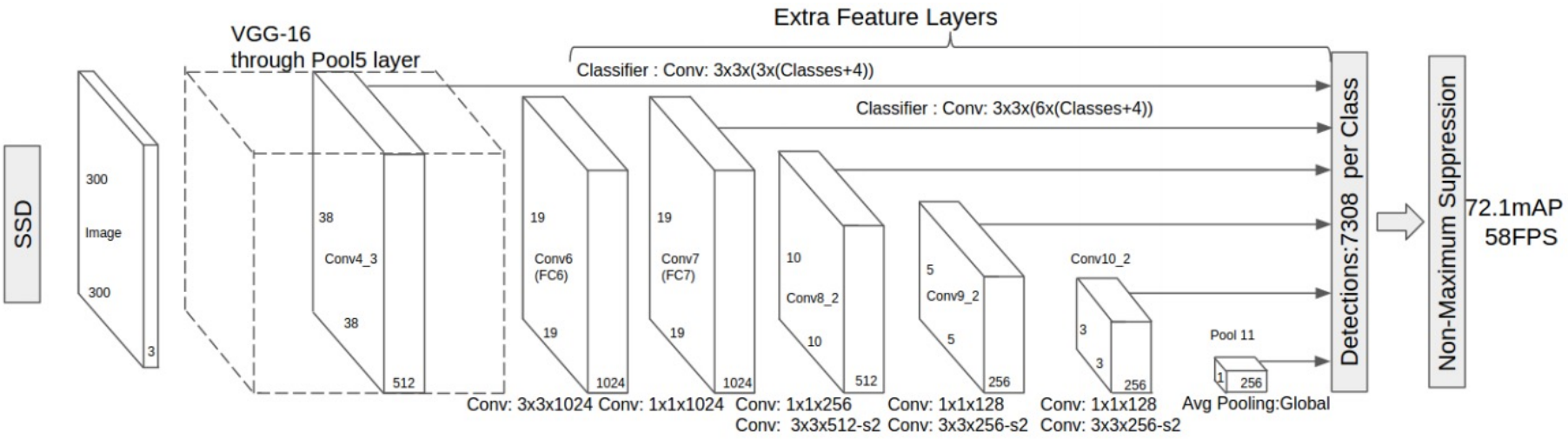
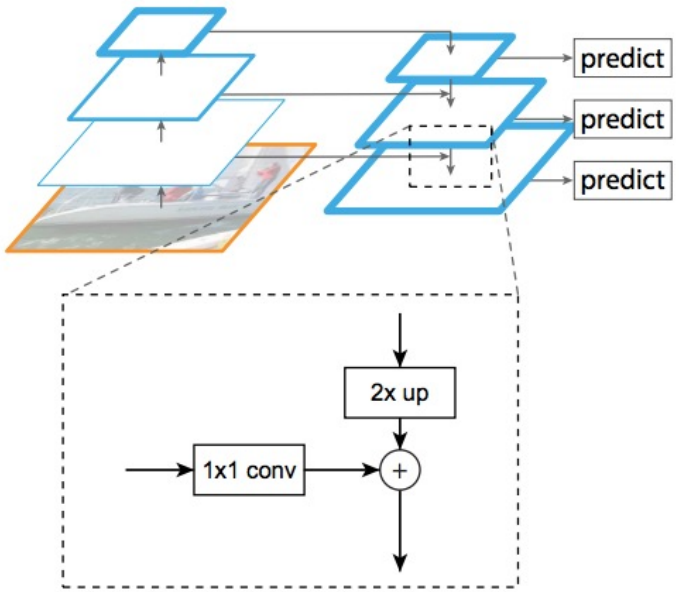



W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. Berg, SSD: Single Shot MultiBox Detector, ECCV 2016

# Feature pyramid networks

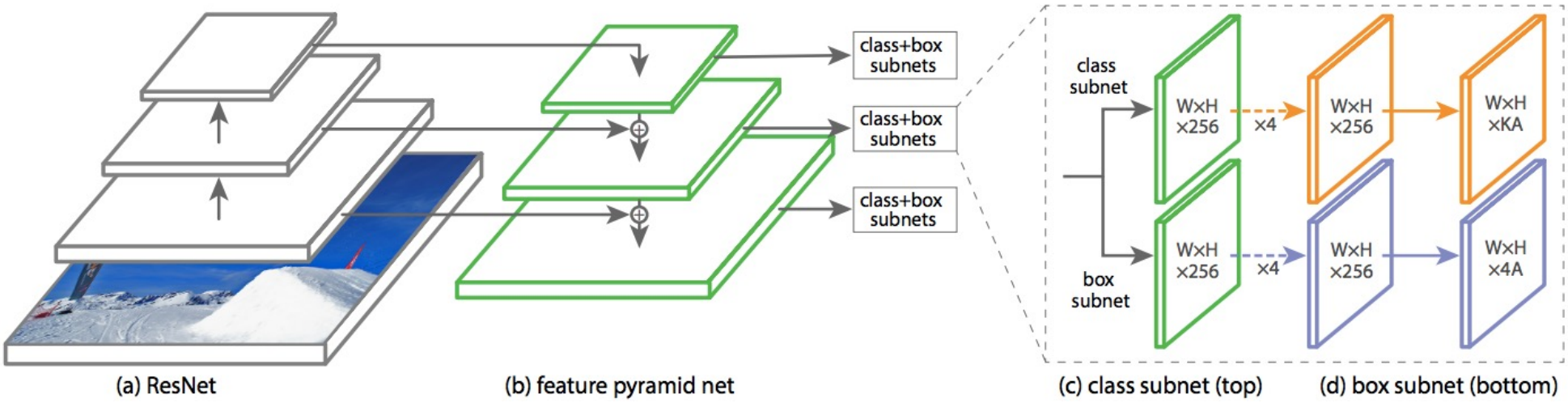
- Improve predictive power of lower-level feature maps by adding contextual information from higher-level feature maps
- Predict different sizes of bounding boxes from different levels of the pyramid (but share parameters of predictors)




T.-Y. Lin, P. Dollar, R. Girshick, K. He, B. Hariharan, and S. Belongie, Feature pyramid networks for object detection, CVPR 2017

# RetinaNet

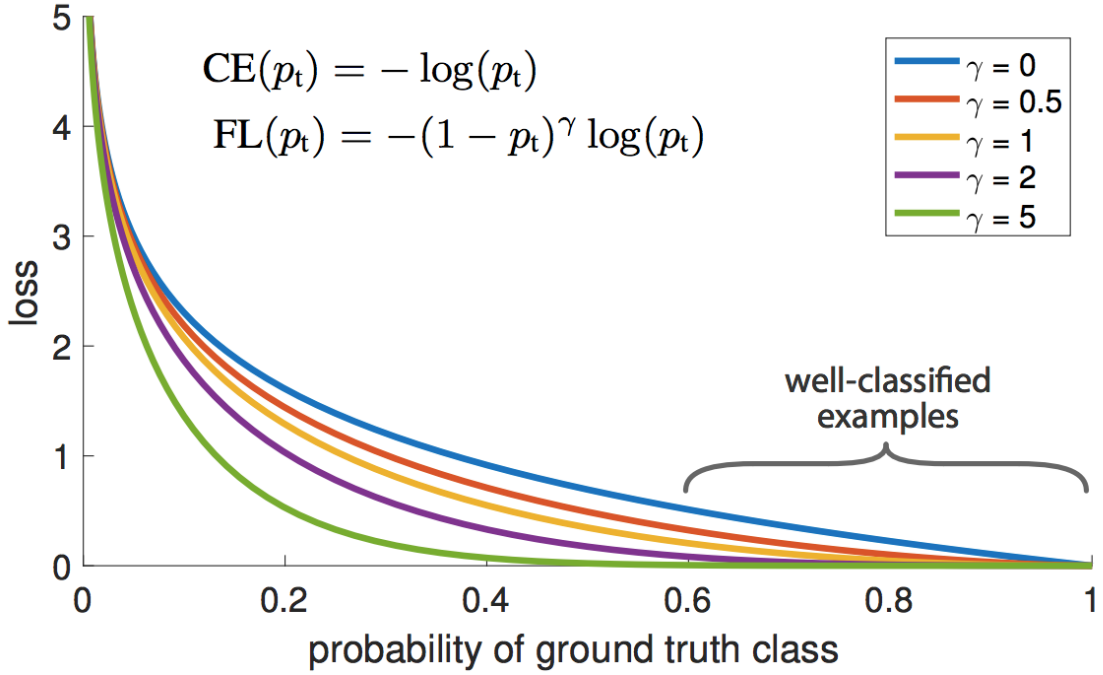
- **Classification subnet**: predict the probability of object at each position for each of *A* anchors and *K* object classes
- **Box subnet**: for each position and each anchor, predict offset to ground truth box (if any)




T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, ICCV 2017

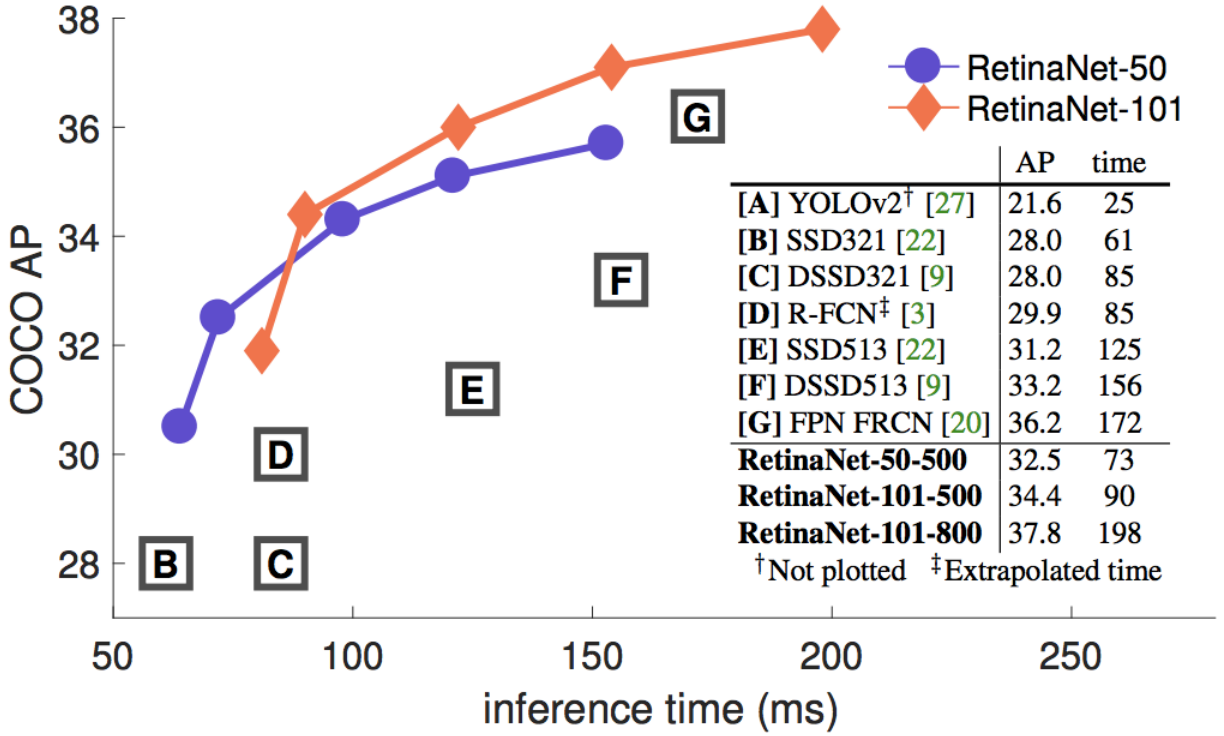
# RetinaNet

- **Focal loss**: down-weight the standard cross-entropy loss for well-classified examples




T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, ICCV 2017

# RetinaNet: Results




| | AP | time |
|---|---|---|
| [A] YOLOv2† [27] | 21.6 | 25 |
| [B] SSD321 [22] | 28.0 | 61 |
| [C] DSSD321 [9] | 28.0 | 85 |
| [D] R-FCN‡ [3] | 29.9 | 85 |
| [E] SSD513 [22] | 31.2 | 125 |
| [F] DSSD513 [9] | 33.2 | 156 |
| [G] FPN FRCN [20] | 36.2 | 172 |
| **RetinaNet-50-500** | 32.5 | 73 |
| **RetinaNet-101-500** | 34.4 | 90 |
| **RetinaNet-101-800** | 37.8 | 198 |

†Not plotted ‡Extrapolated time

T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollar, Focal loss for dense object detection, ICCV 2017

# Fully convolutional one-stage detector (FCOS)

- “Anchor-free” approach

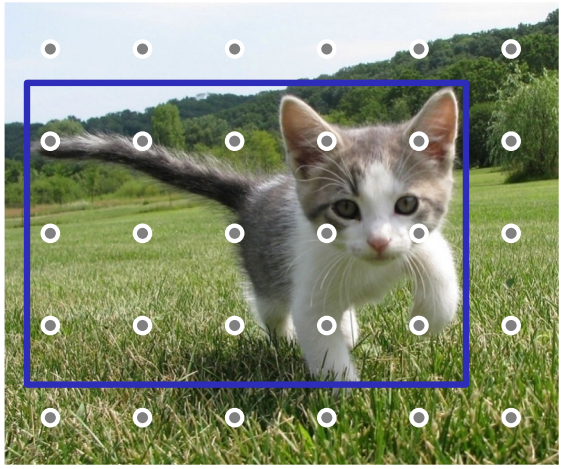


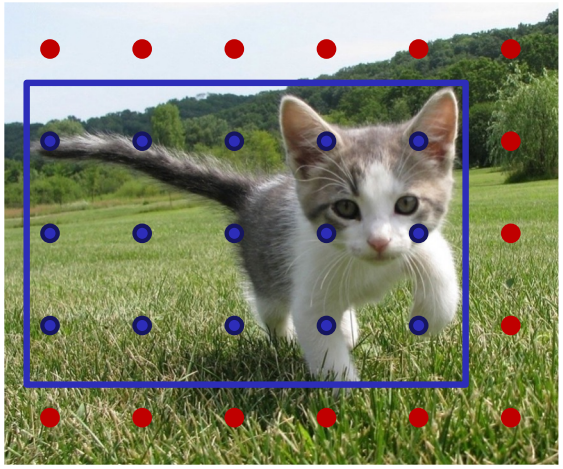
Figure source:
J. Johnson

Run backbone CNN to get features aligned to input image

Tian et al., FCOS: Fully Convolutional One-Stage Object Detection, ICCV 2019

# Fully convolutional one-stage detector (FCOS)

- "Anchor-free" approach



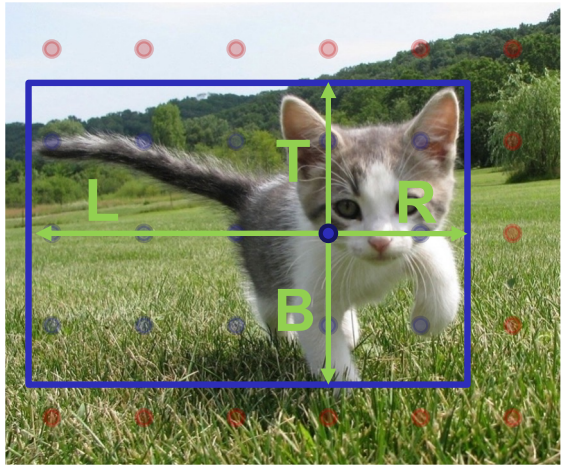
For each class, predict whether location falls inside a GT bounding box

Figure source:
J. Johnson

Tian et al., FCOS: Fully Convolutional One-Stage Object Detection, ICCV 2019

# Fully convolutional one-stage detector (FCOS)

- "Anchor-free" approach




For positive points, also regress distance to left, right, top, and bottom of GT box (with L2 loss)

Figure source: J. Johnson

Tian et al., FCOS: Fully Convolutional One-Stage Object Detection, ICCV 2019

# Fully convolutional one-stage detector (FCOS)

- "Anchor-free" approach

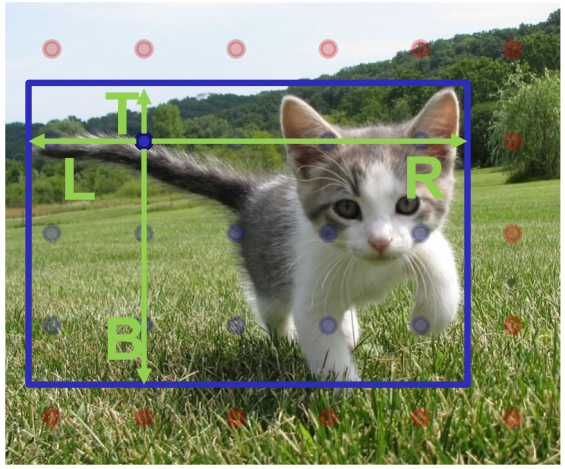


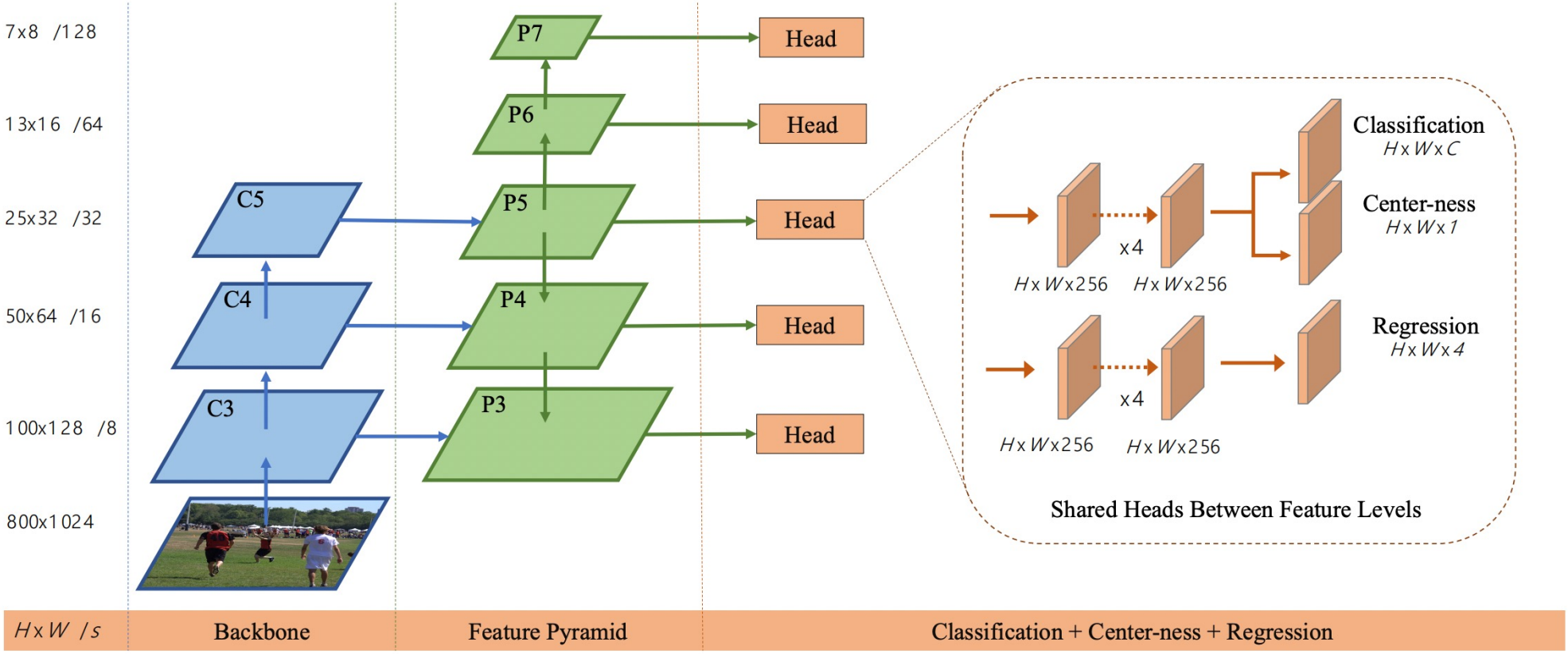

Figure source:
J. Johnson

For positive points, also regress distance to left, right, top, and bottom of GT box (with L2 loss)

Weight detections by "centerness" and confidence, perform NMS

Tian et al., FCOS: Fully Convolutional One-Stage Object Detection, ICCV 2019

# Fully convolutional one-stage detector (FCOS)




Tian et al., FCOS: Fully Convolutional One-Stage Object Detection, ICCV 2019

# Outline

- Task definition and evaluation
- Two-stage detectors
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - Mask R-CNN
- Single-stage and multi-resolution detectors
- Other detectors

# CornerNet

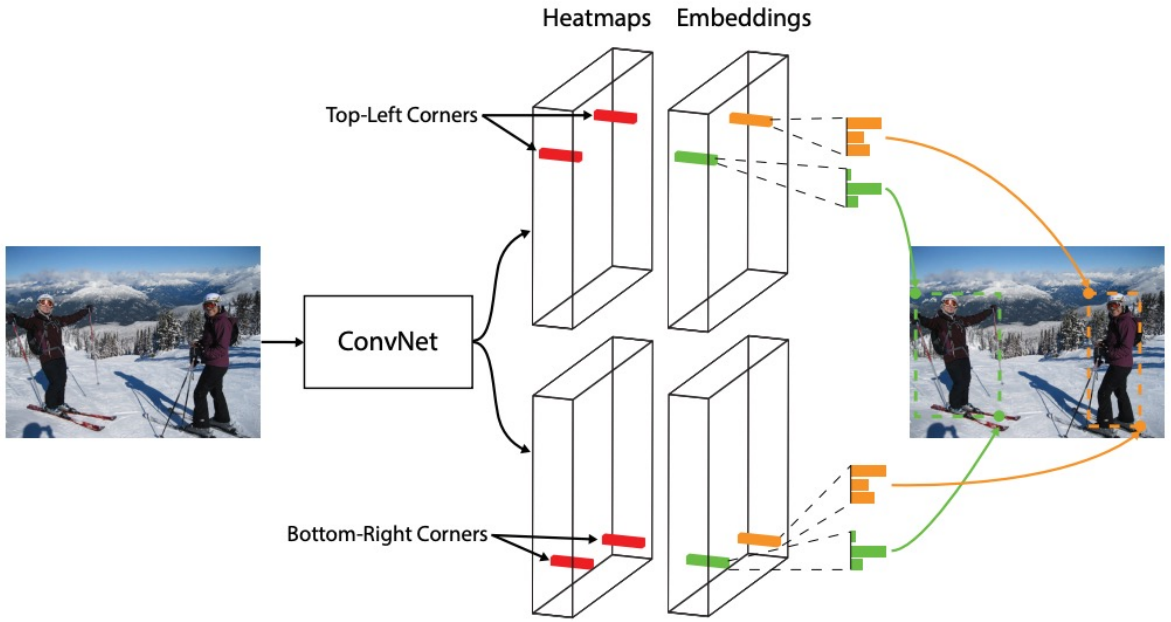
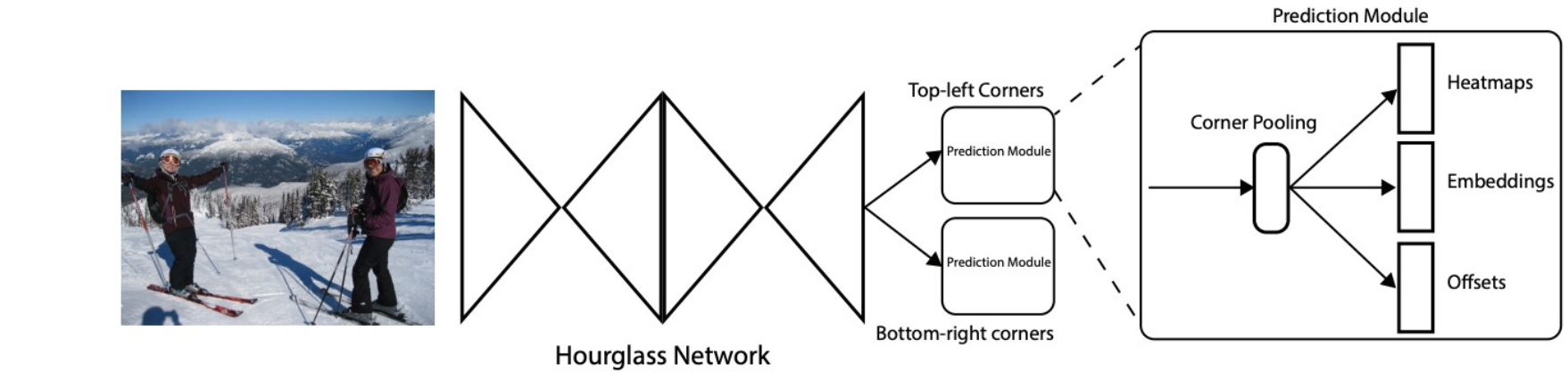
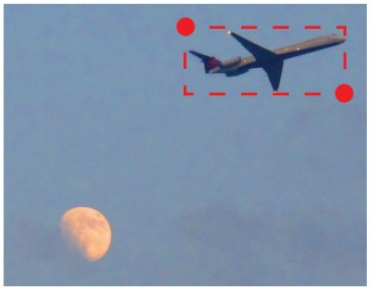



**Fig. 1.** We detect an object as a pair of bounding box corners grouped together. A convolutional network outputs a heatmap for all top-left corners, a heatmap for all bottom-right corners, and an embedding vector for each detected corner. The network is trained to predict similar embeddings for corners that belong to the same object.

H. Law and J. Deng, CornerNet: Detecting Objects as Paired Keypoints, ECCV 2018

# CornerNet



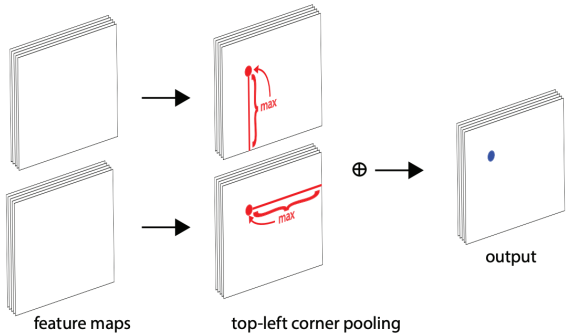
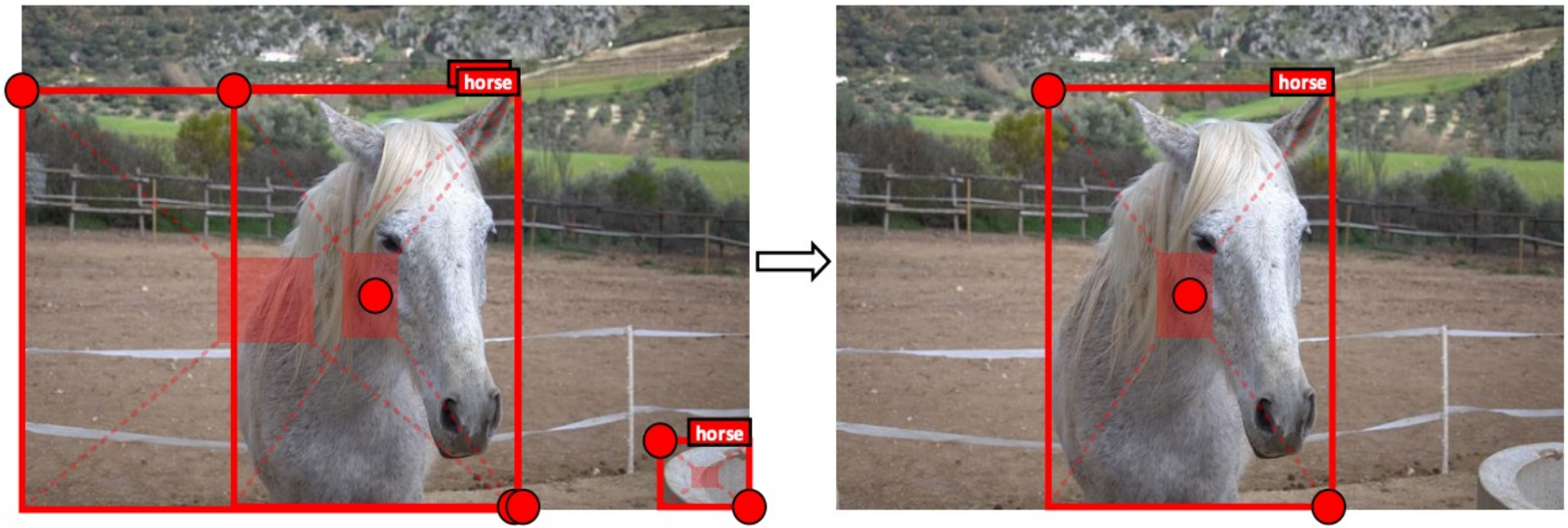



**Fig. 2.** Often there is no local evidence to determine the location of a bounding box corner. We address this issue by proposing a new type of pooling layer.

H. Law and J. Deng, CornerNet: Detecting Objects as Paired Keypoints, ECCV 2018

# CenterNet

- Use an additional center point to verify predictions:




K. Duan et al. CenterNet: Keypoint Triplets for Object Detection, ICCV 2019

# CenterNet

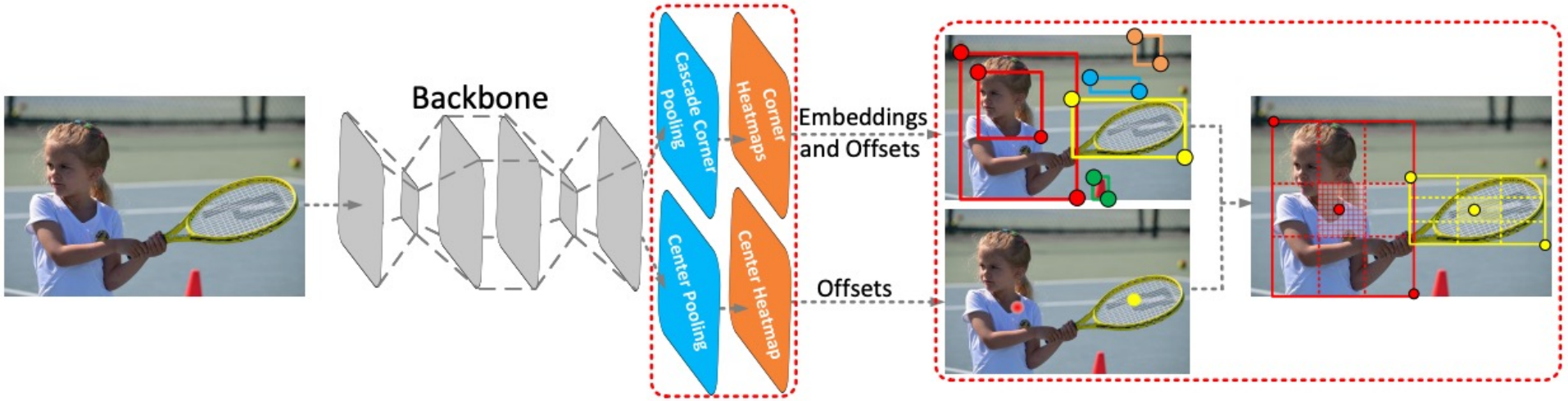



Figure 2: Architecture of CenterNet. A convolutional backbone network applies cascade corner pooling and center pooling to output two corner heatmaps and a center keypoint heatmap, respectively. Similar to CornerNet, a pair of detected corners and the similar embeddings are used to detect a potential bounding box. Then the detected center keypoints are used to determine the final bounding boxes.

K. Duan et al. CenterNet: Keypoint Triplets for Object Detection, ICCV 2019

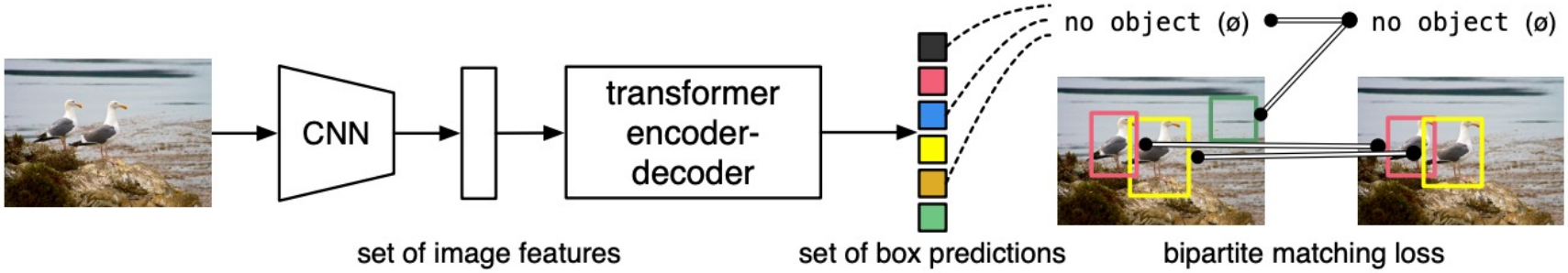
# CenterNet

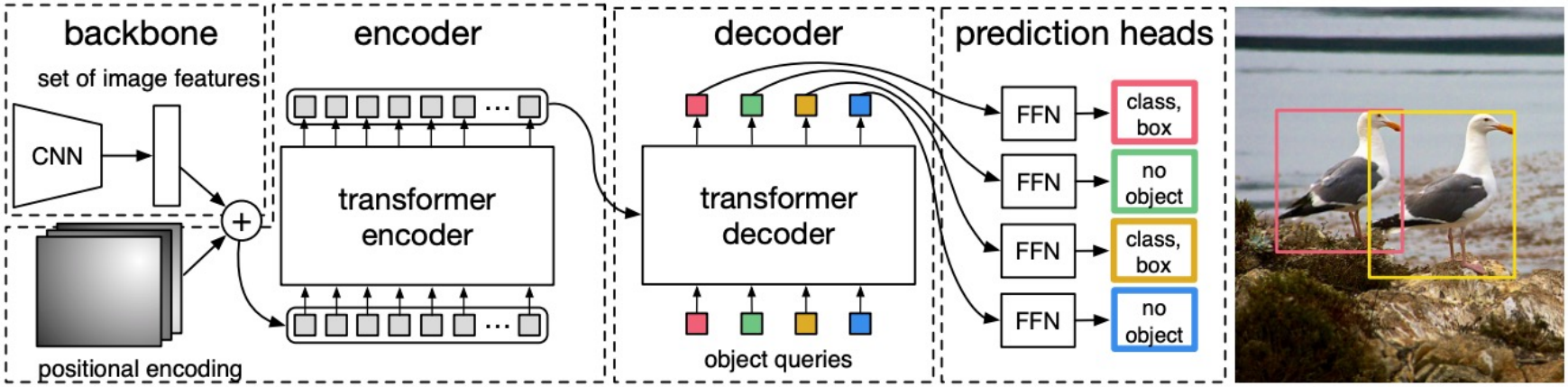
| Method | FD | $FD_5$ | $FD_{25}$ | $FD_{50}$ | $FD_S$ | $FD_M$ | $FD_L$ |
|---|---|---|---|---|---|---|---|
| CornerNet511-52 | 40.4 | 35.2 | 39.4 | 46.7 | 62.5 | 36.9 | 28.0 |
| CenterNet511-52 | **35.1** | **30.7** | **34.2** | **40.8** | **53.0** | **31.3** | **24.4** |
| CornerNet511-104 | 37.8 | 32.7 | 36.8 | 43.8 | 60.3 | 33.2 | 25.1 |
| CenterNet511-104 | **32.4** | **28.2** | **31.6** | **37.5** | **50.7** | **27.1** | **23.0** |

Table 3: Comparison of the false discovery rates (%) of CornerNet and CenterNet on the MS-COCO validation dataset. The results suggest that CenterNet avoids a large number of incorrect bounding boxes, especially for small incorrect bounding boxes.

K. Duan et al. CenterNet: Keypoint Triplets for Object Detection, ICCV 2019

# Detection Transformer (DETR)




N. Carion et al., End-to-end object detection with transformers, ECCV 2020